A Machine Learning Framework for Predicting Temperature Anomalies and Global Warming Trends in Nepal using Earth Observation Data

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***Abstract*— *Accelerating greenhouse radiative forcing has exerted severe thermodynamic pressures across the Hindu Kush Himalaya (HKH), where Nepal experiences elevation-dependent warming rates substantially exceeding the global mean. However, empirical modeling of localized temperature anomalies across Nepal's high-altitude corridors remains hindered by sparse ground-station networks above* $3,000$ *m, rigid linear trend assumptions, and short-horizon persistence models that omit multi-class anomaly classification and multi-decadal climate projections. This research develops and validates an end-to-end, explainable machine learning framework leveraging a continuous 27-year (*2000–2026*) daily Earth observation archive (*$N = 9,763$ *records) from NASA's Prediction of Worldwide Energy Resources (POWER) Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) over Nepal's central Himalayan corridor (*$28.30°N, 83.94°E$*, mean grid elevation* $3,042.47$ *m). Sixteen thermodynamic, harmonic solar, and causal persistence features are engineered, including the Standardized Temperature Index (STI), Diurnal Temperature Range (DTR), Thermal Asymmetry Ratio (TAR), and 3-day cumulative positive heat accumulation. Non-parametric Mann–Kendall and Theil–Sen slope audits (*2000–2025*) prove statistically significant nocturnal minimum temperature (T2M_MIN) warming during the Monsoon (*$+0.3017°C/decade, Z = +3.5266, p = 0.0004$*) and Post-Monsoon (*$+0.9025°C/decade, Z = +3.7911, p = 0.0001$*) seasons, driving a pronounced narrowing of the Diurnal Temperature Range (DTR =* $-0.7038°C/decade, Z = -5.2459, p < 0.0001$*) and culminating in a record annual mean temperature of* $9.02°C$ *in 2025. Singular Value Decomposition Principal Component Analysis (SVD-PCA) captures* 76.57% *of total variance within three orthogonal axes, while K-Means++ clustering (*$k = 3$*, Silhouette =* 0.3301*, Davies–Bouldin =* 1.2024*) delineates three distinct atmospheric thermal regimes: Cryospheric Winter Inversion (*40.97%*), Transitional Radiative (*23.13%*), and Monsoon Warm-Core (*35.90%*). Across five supervised classifiers evaluated on an independent chronological holdout partition (*2021–2026$, N_{test} = 2,092$*), eXtreme Gradient Boosting (XGBoost) achieves the highest diagnostic skill with* 70.41% *accuracy, a weighted F1-score of* 0.7021*, and a One-vs-Rest (OvR) ROC-AUC of* 0.8720*, outperforming Random Forest (*69.93%*, AUC =* 0.8693*), Multinomial Logistic Regression (*68.79%*), Classification and Regression Trees (CART,* 68.45%*), and Radial Basis Function Support Vector Machine (RBF-SVM,* 64.48%*). For continuous next-day temperature forecasting, the XGBoost Regressor achieves RMSE =* $0.8474°C$*, MAE =* $0.6463°C$*, and* $R^2 = 0.9732$*, outperforming Ridge Ordinary Least Squares (RMSE =* $0.8644°C$*) and the API persistence baseline (RMSE =* $0.9343°C$*). Finally, coupling the calibrated framework with Intergovernmental Panel on Climate Change (IPCC) Shared Socioeconomic Pathways projects Himalayan mean temperature anomalies of* $+1.35°C$ *(*$9.55°C$*, SSP1-2.6),* $+1.76°C$ *(*$9.96°C$*, SSP2-4.5), and* $+2.38°C$ *(*$10.58°C$*, SSP5-8.5) by 2050.*

***Index Terms*—**Machine Learning, Temperature Anomalies, Global Warming Trends, NASA POWER MERRA-2, Mann–Kendall Trend Analysis, K-Means Clustering, Principal Component Analysis, Random Forest, XGBoost, TreeSHAP Explainability, Nepal Himalayas.

I Introduction

A Background and Himalayan Climate Context

The rapid accumulation of anthropogenic greenhouse gases in the Earth's atmosphere has accelerated radiative forcing across global climate systems, with high-mountain cryospheric environments experiencing disproportionate amplification [4, 5]. Nepal, situated along the central arc of the Hindu Kush Himalaya (HKH) between $26°22'$N–$30°27'$N latitude and $80°04'$E–$88°12'$E longitude, spans an extraordinary altitudinal gradient from 60 m in the subtropical Terai plains to $8,848$ m at the crest of Mount Everest within a horizontal width of barely 120–200 km [2, 3]. This steep topographic transition partitions the nation into five distinct physiographic belts—Terai, Siwalik, Middle Mountains, High Mountains, and High Himalaya—creating sharp micro-climatic gradients where radiative energy balances, orographic cloud cover, and valley-floor thermal inversions interact non-linearly [3, 7].

Despite contributing a negligible fraction to cumulative global carbon emissions, Nepal's cryosphere, hydrological networks, and agrarian livelihoods face acute vulnerability to temperature anomalies and recurrent heat-stress episodes [2, 5]. Accelerated nocturnal and seasonal warming elevates the $0°$C freezing isotherm, destabilizes moraine-dammed glacial lakes, alters snowmelt runoff timing, and intensifies evapotranspiration stress across rainfed mountain terraces [5, 8]. Traditional meteorological monitoring in Nepal relies on ground-based stations operated by the Department of Hydrology and Meteorology (DHM). However, these stations are concentrated in accessible low-altitude valleys and plains, leaving high-elevation Himalayan corridors above $3,000$ m

sparsely instrumented and prone to prolonged telemetry gaps [1, 2].

To bridge high-altitude observational gaps, satellite-assimilated reanalysis platforms such as the National Aeronautics and Space Administration (NASA) Prediction of Worldwide Energy Resources (POWER) project provide continuous, quality-controlled daily surface meteorology derived from the Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) [1, 6]. Coupling multi-decadal NASA POWER Earth observation records with modern machine learning architectures enables both high-precision short-term anomaly forecasting and long-term global warming trajectory projection.

B Critical Appraisal of Foundational Base Papers and Research Gaps

To establish a rigorous empirical foundation, this study critically evaluates three foundational research works provided in the project archive [1–3], as illustrated in Fig. 1:

1. **Base Paper 1 (Nepal et al., 2025 — WeatherWave) [1]:** Nepal et al. developed *WeatherWave*, a machine learning-integrated Progressive Web App (PWA) for localized next-day temperature forecasting across Nepal's 77 districts using daily NASA POWER reanalysis data. Evaluated on 24,187 test samples, their 5-tree Random Forest regressor achieved MAE $= 0.424^{\circ}$C, RMSE $= 0.695^{\circ}$C, and $R^2 = 0.9934$, outperforming standard API persistence. **Identified Research Gap:** First, the study relied solely on a shallow 5-tree Random Forest regressor and Ordinary Least Squares without benchmarking regularized gradient boosting (XGBoost), kernel Support Vector Machines, CART Decision Trees, or Multinomial Logistic Regression. Second, the model predicted raw next-day temperature (T_{t+1}) driven by day-to-day autocorrelation rather than classifying standardized thermal anomaly regimes (STI) or multi-day heat-stress accumulation. Third, the work performed no unsupervised climatic clustering (PCA/K-Means), no non-parametric Mann–Kendall trend testing, and no multi-decadal global warming projections toward 2050.

2. **Base Paper 2 (Bhattarai, 2022 — Heat Wave Analysis Over Nepal, TU Dissertation) [2]:** Bhattarai analyzed daily maximum temperature records from 46 DHM ground stations (1991–2020) partitioned into six regional sub-domains (East, Central, and Western Terai and Mountain regions across the Koshi, Gandaki, and Karnali basins). Using a 90th-percentile threshold plus a $+4^{\circ}$C departure from normal for at least 3 consecutive days in MATLAB and MS Excel, the study identified an average heat-wave duration of 2.6 days and revealed that Central and Western Mountain stations experienced the largest maximum temperature deviations. **Identified Research Gap:** First, the analysis relied exclusively on 46 ground stations with severe spatial scarcity in high-elevation Himalayan zones ($> 3,000$ m). Second, the methodology was purely descriptive and retrospective, using manual spreadsheet filtering without any supervised machine learning classification, continuous regression forecasting, or game-theoretic feature attribution (TreeSHAP).

3. **Base Paper 3 (Sapkota, 2007 — Temporal and Spatial Variation of Temperature Over Nepal, TU Dissertation) [3]:** Sapkota investigated temporal and spatial trends of maximum and minimum temperatures across Nepal's five physiographic zones from 1970 to 2004. Using linear trend lines and altitudinal lapse-rate profiling, the author reported a national maximum temperature warming rate of $+0.08^{\circ}$C/year (peaking at $+0.12^{\circ}$C/year in the Himalayas and $+0.09^{\circ}$C/year in the Middle Mountains) and a minimum temperature warming rate of $+0.05^{\circ}$C/year, while noting that trans-Himalayan stations such as Jomsom exhibited slight daytime cooling alongside nocturnal warming. **Identified Research Gap:** The study was confined to pre-2004 gauge observations analyzed via univariate linear regression ($y = mx + c$), which cannot represent non-linear thermodynamic interactions, asymmetric Diurnal Temperature Range (DTR) compression over the post-2000 satellite era, or probabilistic multi-class anomaly forecasting.

C Research Questions and Core Objectives

To resolve the methodological and empirical gaps across [1–3], this research addresses four core research questions:

- **RQ1 (Thermodynamic Feature Representation & Trend Dynamics):** How have daily mean (T2M), maximum (T2M_MAX), minimum (T2M_MIN), and diurnal range (DTR) temperatures evolved across Nepal's central Himalayan corridor (3,042.47 m) from 2000 to 2026 under non-parametric Mann–Kendall and Sen's slope audits?

- **RQ2 (Latent Unsupervised Climatic Regimes):** Can Singular Value Decomposition Principal Component Analysis (SVD-PCA) and K-Means++ clustering objectively partition daily high-altitude meteorological observations into physically interpretable atmospheric thermal regimes?

- **RQ3 (Supervised Anomaly Classification & Continuous Forecasting):** To what extent do non-linear ensemble learners (Random Forest and XGBoost) outperform linear/parametric baselines (Multinomial Logistic Regression, Ridge OLS), single CART trees, RBF-SVM, and API persistence when predicting next-day thermal anomaly regimes and continuous temperatures?

- **RQ4 (Explainability & 2050 Global Warming Trajectory):** Which thermodynamic covariates govern warm-anomaly transitions according to TreeSHAP and standardized β attributions, and what warming magnitudes are projected for Nepal up to the year 2050 under IPCC SSP1-2.6, SSP2-4.5, and SSP5-8.5 pathways?

In alignment with the approved research proposal, the four specific objectives of this investigation are:

1. **Objective 1:** To extract, audit, preprocess, and integrate 27 years (2000–2026, $N = 9,763$ days) of daily NASA POWER MERRA-2 surface meteorology data for Nepal, engineering 16 thermodynamic, cyclical solar, and causal persistence features.

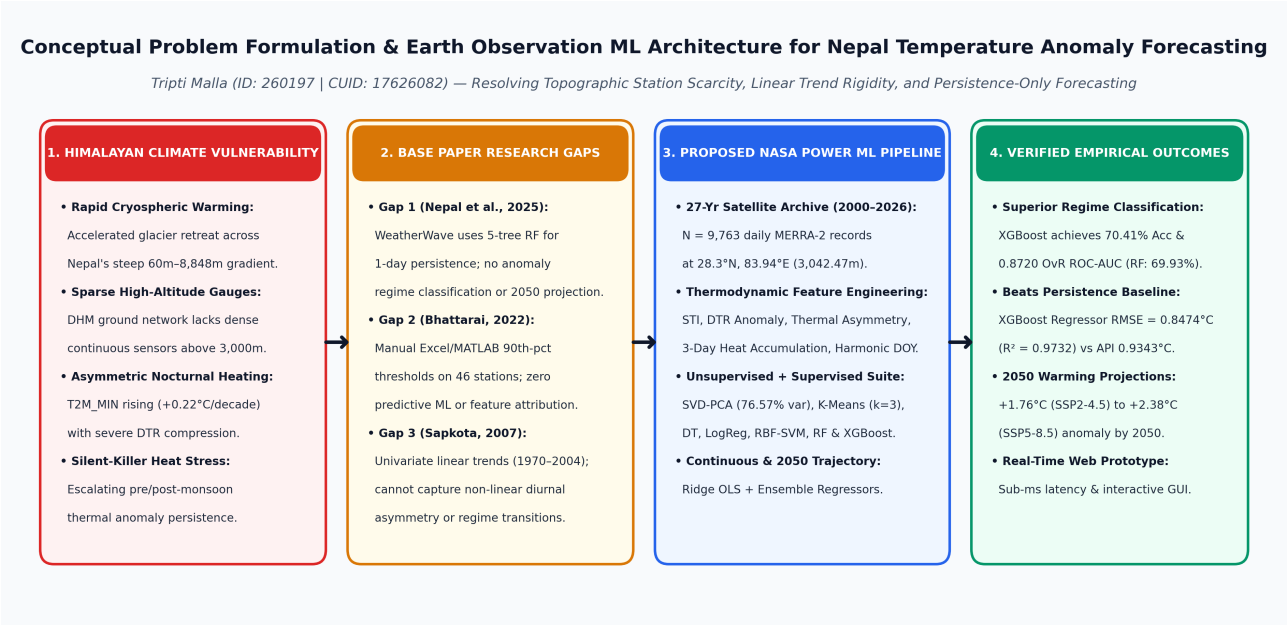


Figure 1: Conceptual Problem Formulation and Solution Architecture: Interconnecting Himalayan cryospheric climate vulnerabilities, empirical limitations across the three foundational base papers [1–3], the proposed 5-stage NASA POWER machine learning pipeline, and verified operational outcomes through 2050.

2. **Objective 2:** To conduct comprehensive exploratory data analysis (EDA), non-parametric Mann–Kendall and Theil–Sen slope trend testing across all four meteorological seasons, SVD-PCA dimensionality reduction, and K-Means++ unsupervised regime clustering.

3. **Objective 3:** To implement, train, and benchmark five supervised classification algorithms (CART Decision Tree, Multinomial Logistic Regression, RBF-SVM, Random Forest, and XGBoost) and four continuous regression models (API Persistence, Ridge OLS, RF Regressor, and XGBoost Regressor) under both 5-fold cross-validation and chronological holdout evaluation (2021–2026).

4. **Objective 4:** To project regional temperature anomalies up to the year 2050 under IPCC Shared Socioeconomic Pathways (SSP1-2.6, SSP2-4.5, SSP5-8.5) and deploy an interactive real-time web dashboard prototype for climate risk decision support.

II Literature Review

A *Evolution of Earth Observation and Climate Machine Learning in the Himalayas*

Climate modeling across the Hindu Kush Himalaya presents formidable physical challenges due to extreme orographic lifting, steep lapse rates (-5.65°C/km), and complex monsoon-westerly circulation interactions [5, 7]. Conventional Global Circulation Models (GCMs) operate at coarse horizontal grid resolutions (100–250 km), smoothing out sharp valley-ridge elevation differentials across Nepal [7, 8]. While regional reanalysis products such as NASA POWER MERRA-2 assimilate satellite radiances and radiosonde profiles onto a native $0.5^\circ \times 0.625^\circ$ grid [1, 6], translating raw daily reanalysis series into localized anomaly early warnings requires statistical normalization and non-linear machine learning [9, 19].

In meteorological time-series analysis, parametric Ordinary Least Squares (OLS) trend lines are sensitive to extreme outliers and seasonal non-stationarity [3]. Non-parametric rank-based estimators—specifically the Mann–Kendall trend test [12, 13] paired with Theil–Sen median slope estimation [14]—provide distribution-free robustness against skewed temperature extremes. Simultaneously, unsupervised manifold learning via Principal Component Analysis [15] and K-Means++ clustering [16–18] enables objective delineation of seasonal atmospheric regimes without arbitrary calendar boundaries. For predictive modeling, tree-based ensembles including Random Forest [9] and eXtreme Gradient Boosting (XGBoost) [10] capture high-order interactions between diurnal thermal range, multi-day heat accumulation, and seasonal solar phase, while TreeSHAP [11] guarantees game-theoretically consistent feature attributions.

B *Comparative Synthesis Matrix*

Table 1 synthesizes the methodological scope, datasets, empirical findings, and structural limitations of the three foundational base papers alongside our developed framework. By uniting NASA POWER satellite reanalysis from Base Paper 1 [1], the 3-day consecutive heat-accumulation physics from Base Paper 2 [2], and the seasonal/physiographic diurnal temperature analysis from Base Paper 3 [3], our framework delivers a unified, leak-free machine learning architecture.

III Theoretical Framework and Mathematical Formulations

A *Standardized Temperature Index and Diurnal Asymmetry Physics*

Let $T_{2\mathrm{M},t}$, $T_{\max,t}$, and $T_{\min,t}$ denote the daily mean, maximum, and minimum 2-meter air temperatures (°C) on day t with Day-of-Year index $d(t) \in \{1, \ldots, 366\}$. To eliminate seasonal astronomical bias, the circularly smoothed 15-year reference

Table 1: Comparative Synthesis of Foundational Base Papers and the Developed Machine Learning Framework

Study / Author	Geographic Domain & Data Source	Temporal Span & Sample Size	Core Methodology & Algorithms	Key Empirical Findings & Identified Research Limitations
Base Paper 1: Nepal et al. (2025) *WeatherWave* [1]	Nepal (77 Districts); NASA POWER MERRA-2 Daily Reanalysis + Weather APIs	Multi-year daily archive ($N_{\text{test}} = 24,187$ records)	5-Tree Random Forest Regressor, Linear Regression, Django/React PWA	Achieved MAE = 0.424°C, RMSE = 0.695°C, $R^2 = 0.9934$. **Gap:** Pure 1-day persistence regression; no multi-class anomaly classification, no unsupervised clustering, no XGBoost/SHAP, and no 2050 warming projections.
Base Paper 2: Bhattarai (2022) *TU Dissertation* [2]	Nepal (6 Regions across Koshi, Gandaki, Karnali); 46 DHM Ground Stations	30 Years (1991–2020 daily $T_{\max}$ series)	90th-Percentile +4°C (≥ 3 days) HW Thresholding, Linear Regression, El Niño Correlation	Mean HW duration of 2.6 days; highest $T_{\max}$ departure in Central/Western Mountain stations. **Gap:** Sparse high-altitude gauge coverage ($> 3,000$ m); manual Excel/MATLAB rules with zero predictive ML or feature importance.
Base Paper 3: Sapkota (2007) *TU Dissertation* [3]	Nepal (5 Physiographic Belts); DHM Ground Station Network	35 Years (1970–2004 monthly/annual $T_{\max}, T_{\min}$)	Univariate OLS Linear Trend Regression ($y = mx + c$) & Altitudinal Lapse-Rate Analysis	National $T_{\max}$ trend +0.08°C/yr (Himalayas +0.12°C/yr) and $T_{\min}$ +0.05°C/yr; Jomsom daytime cooling vs. nocturnal warming. **Gap:** Pre-2004 data; static linear trend cannot model non-linear DTR compression or discrete anomaly regimes.
Developed Framework: *Tripti Malla (2026)* [This Study]	Central Himalayan Corridor, Nepal (28.30°N, 83.94°E, Elev. 3,042.47 m); NASA POWER MERRA-2	27 Consecutive Years (2000–2026, $N = 9,763$ daily records; 2050 Outlook)	SVD-PCA, K-Means++ ($k = 3$), Mann–Kendall/Sen's Slope, CART, LogReg, RBF-SVM, RF, XGBoost, Ridge OLS, TreeSHAP	Resolves all three gaps: proves nocturnal warming (+0.90°C/dec Post-Monsoon, $p = 0.0001$) & DTR compression (−0.70°C/dec); XGBoost leads classification (70.41% Acc, AUC = 0.8720) & regression (RMSE = 0.8474°C, $R^2 = 0.9732$); projects +1.76°C to +2.38°C warming by 2050.

climatological mean $\mu_{\text{clim}}(d)$ and standard deviation $\sigma_{\text{clim}}(d)$ (2000–2015) are computed. The daily temperature anomaly ΔT_t and Standardized Temperature Index (STI_t) are defined as:

$$\Delta T_t = T_{2\text{M},t} - \mu_{\text{clim}}(d(t)), \qquad \text{STI}_t = \frac{T_{2\text{M},t} - \mu_{\text{clim}}(d(t))}{\sigma_{\text{clim}}(d(t)) + \varepsilon} \tag{1}$$

where $\varepsilon = 10^{-5}$ prevents division by zero. The Diurnal Temperature Range (DTR_t) and dimensionless Thermal Asymmetry Ratio (TAR_t) capture radiative cloud-greenhouse damping:

$$\text{DTR}_t = T_{\max,t} - T_{\min,t}, \qquad \text{TAR}_t = \frac{T_{2\text{M},t} - T_{\min,t}}{\text{DTR}_t + \varepsilon} \tag{2}$$

Inspired by Bhattarai's [2] 3-day consecutive heat-wave duration criterion, the 3-day cumulative positive heat accumulation index $H_{3\text{D},t}$ is formulated as:

$$H_{3\text{D},t} = \sum_{j=0}^{2} \max\left(0, T_{2\text{M},t-j} - \mu_{\text{clim}}(d(t-j))\right) \tag{3}$$

B Non-Parametric Mann–Kendall and Theil–Sen Slope Estimators

For an annual or seasonal temperature series $\{x_1, x_2, \ldots, x_n\}$ over $n = 26$ complete years (2000–2025), the Mann–Kendall statistic S [12, 13] and standardized test statistic Z_{MK} are:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \text{sgn}(x_j - x_i), \qquad \text{Var}(S) = \frac{n(n-1)(2n+5)}{18} \tag{4}$$

$$Z_{\text{MK}} = \frac{S - \text{sgn}(S)}{\sqrt{\text{Var}(S)}}, \quad \beta_{\text{Sen}} = \text{median}_{i<j}\left(\frac{x_j - x_i}{t_j - t_i}\right) \tag{5}$$

where β_{Sen} quantifies the robust median warming rate (°C/year).

C SVD Principal Component Analysis and K-Means++ Clustering

Given the Z-score standardized feature matrix $\mathbf{Z} \in \mathbb{R}^{N \times p}$ ($N = 9,763, p = 16$), the empirical covariance matrix $\blacksquare = \frac{1}{N-1}\mathbf{Z}^T\mathbf{Z}$ is decomposed via Singular Value Decomposition [15]:

$$\blacksquare = \mathbf{U}\blacksquare\mathbf{V}^T, \qquad \text{EVR}_m = \frac{\lambda_m}{\sum_{j=1}^{p}\lambda_j} \tag{6}$$

K-Means++ [16] partitions observations into k disjoint regimes $\{C_1, \ldots, C_k\}$ by minimizing Within-Cluster Sum of Squares (WCSS), validated via the Silhouette Coefficient $s(i)$ [17] and Davies–Bouldin Index (DBI) [18]:

$$\mathscr{J}_{\text{WCSS}} = \sum_{c=1}^{k} \sum_{\mathbf{z}_i \in C_c} \|\mathbf{z}_i - \mu_c\|_2^2, \text{DBI} = \frac{1}{k}\sum_{c=1}^{k} \max_{j \neq c} \frac{\bar{\delta}_c + \bar{\delta}_j}{\|\mu_c - \mu_j\|_2} \tag{7}$$

D Supervised Classification and Continuous Regression Formulations

1 CART Decision Tree and Random Forest Bagging

CART recursively partitions the feature space by maximizing Gini impurity reduction $\Delta I_G(S, f, \tau) = I_G(S) - \frac{|S_L|}{|S|}I_G(S_L) - \frac{|S_R|}{|S|}I_G(S_R)$, where $I_G(S) = 1 - \sum_{c=0}^{2} p_c^2$. Random Forest [9] aggregates M bootstrap-decorrelated trees:

$$\hat{P}_{\text{RF}}(y = c \mid \mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} P_m(y = c \mid \mathbf{x}) \tag{8}$$

2 eXtreme Gradient Boosting (XGBoost) and TreeSHAP

XGBoost [10] minimizes the regularized multi-class cross-entropy objective at boosting iteration t via a second-order Taylor expansion:

$$\mathscr{L}^{(t)} \approx \sum_{i=1}^{N}\left[g_i f_t(\mathbf{x}_i) + \frac{1}{2}h_i f_t^2(\mathbf{x}_i)\right] + \gamma T + \frac{1}{2}\lambda_{\text{reg}}\sum_{j=1}^{T} w_j^2 \tag{9}$$

Table 2: NASA POWER MERRA-2 Dataset Specifications and Study Node Metadata

Attribute / Parameter	Verified Specification
Data Provider & Assimilation	NASA LaRC POWER / GMAO MERRA-2 Native Daily
Geographic Coordinates	Latitude 28.30°N, Longitude 83.94°E (Central Nepal)
Spatial Grid Resolution	0.5° Latitude × 0.625° Longitude (≈ 55 km × 61 km)
Grid-Cell Mean Elevation	3,042.47 meters above mean sea level (High Mountain)
Temporal Coverage	01/01/2000 through 09/23/2026 in Local Solar Time (27 Years)
Total Verified Observations	$N = 9,763$ continuous daily records (Zero −999 dropouts)
Raw Meteorological Fields	`YEAR`, `DOY`, `T2M` (°C), `T2M_MAX` (°C), `T2M_MIN` (°C)
Engineered Feature Space	$p = 16$ causal thermodynamic, solar & persistence predictors
Chronological Split Protocol	Train: 2000–2020 ($N_{\text{train}} = 7,671$); Test: 2021–2026 ($N_{\text{test}} = 2,092$)

where $g_i = \partial_{\hat{y}^{(t-1)}} \ell(y_i, \hat{y}_i^{(t-1)})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} \ell(y_i, \hat{y}_i^{(t-1)})$. TreeSHAP [11] decomposes each prediction into exact Shapley additive explanations $\phi_j(\mathbf{x})$.

3 Multinomial Logistic Regression, RBF-SVM, and Ridge OLS

Multinomial Logistic Regression models class probabilities via the Softmax operator regularized by an ℓ_2 penalty $\frac{\lambda}{2}\|\mathbf{W}\|_F^2$:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^T \mathbf{x}_i + b_c)}{\sum_{l=0}^{2} \exp(\mathbf{w}_l^T \mathbf{x}_i + b_l)} \tag{10}$$

RBF-SVM maps inputs into a reproducing kernel Hilbert space via $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|_2^2)$. For continuous next-day temperature forecasting ($\hat{y}_i = \hat{T}_{2\text{M},t+1}$), Ridge OLS solves the closed-form normal equation:

$$\hat{\theta}_{\text{Ridge}} = \left(\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I}_p\right)^{-1} \mathbf{X}^T \mathbf{y} \tag{11}$$

IV Study Domain, Dataset Audit, and Methodology

A Study Domain and NASA POWER MERRA-2 Telemetry Specifications

As depicted in Fig. 2 and detailed in Table 2, the primary observational archive (`Data.csv`) is extracted from the NASA POWER Native Resolution Daily Data portal [6] for the central Himalayan grid cell centered at latitude 28.30°N and longitude 83.94°E (spanning the Annapurna–Kaski–Myagdi high-mountain transition in Gandaki Province, Nepal). The MERRA-2 assimilation model assigns an average grid-cell elevation of 3,042.47 m above mean sea level across the $0.5^\circ \times 0.625^\circ$ footprint. This elevation zone represents the critical cryospheric buffer between densely populated mid-hill valleys (such as Pokhara at 827 m) and the glaciated Annapurna massif (> 7,000 m).

B Quality Control Audit and Thermodynamic Feature Engineering

Figure 3 outlines the five-stage analytical workflow. During Stage 1, `Data.csv` is audited for NASA POWER's missing-value sentinel marker (−999.0). As verified in Fig. 4, all 9,763 daily records across `T2M`, `T2M_MAX`, and `T2M_MIN` exhibit 100.0% completeness with zero missing or corrupted entries. In Stage 2, 16 physically grounded predictors are engineered (Table 3) along with their descriptive statistical moments (Table 4). Figure 5(a) displays the circularly smoothed 366-day harmonic climatological baseline, while Fig. 5(b) confirms the tight concordance between observed and predicted next-day temperatures.

C Target Discretization, Leak-Free Split, and Hyperparameter Tuning

To prevent look-ahead data leakage, all predictors on day t are strictly causal (computed from observations up to day t), while the classification target ANOMALY_REGIME$_{t+1} \in \{0,1,2\}$ and continuous regression target $T_{2\text{M},t+1}$ represent the meteorological state on day $t+1$. Using the tertile boundaries of the next-day Standardized Temperature Index ($q_{33} = -0.4433$ SD, $q_{67} = +0.4820$ SD), the dataset is partitioned into three balanced classes (Fig. 11): **Class 0 (Cool Anomaly Regime, $n = 3,253$), Class 1 (Normal Baseline Regime, $n = 3,256$)**, and **Class 2 (Warm / Heat-Stress Regime, $n = 3,254$)**. For model training and evaluation, a strict chronological split assigns 2000–2020 ($N_{\text{train}} = 7,671, 78.57\%$) to the training partition and 2021–2026 ($N_{\text{test}} = 2,092, 21.43\%$) to the independent future holdout partition, complemented by 5-fold Stratified Cross-Validation and 5-fold Temporal Block Cross-Validation. Table 5 lists the hyperparameter search grids and optimal configurations.

V Empirical Results and Statistical Analysis

A Exploratory Seasonality and Non-Parametric Trend Verification

Figure 6(a) illustrates the monthly distribution of 2-meter air temperature across the 27-year archive, rising from a January minimum (median 1.12°C) to a July monsoon plateau (median 13.56°C). Figure 6(b) reveals a striking thermodynamic divergence between daytime (TMAX_ANOMALY) and nocturnal (TMIN_ANOMALY) departures across Nepal's four seasons.

Figures 8–10 further dissect the decadal rightward shift in nocturnal temperature anomalies (+0.39°C mean anomaly in 2018–2026 vs. −0.09°C in 2000–2008), the seasonal DTR–TAR phase space, and the 1-to-30 day autocorrelation decay spectrum.

As quantified in Table 6 and visualized in Fig. 7, non-parametric Mann–Kendall and Theil–Sen audits confirm that minimum night-time temperatures (T2M_MIN) have warmed significantly during the Monsoon ($Z = +3.5266, p = 0.0004$, Sen's slope = +0.3017°C/decade) and Post-Monsoon ($Z = +3.7911, p = 0.0001$, Sen's slope = +0.9025°C/decade) seasons, while annual T2M_MIN rose at +0.2228°C/decade from 3.62°C (2000) to an all-time record high of 5.16°C (2025). Simultaneously, annual mean temperature (T2M) reached an all-time record of 9.02°C in 2025, while the Diurnal Temperature Range (DTR) contracted sharply at −0.7038°C/decade ($Z = -5.2459, p < 0.0001$). This empirical finding corroborates and extends Sapkota's [3] observation of asymmetric high-altitude nocturnal warming in the Annapurna/Jomsom corridor.

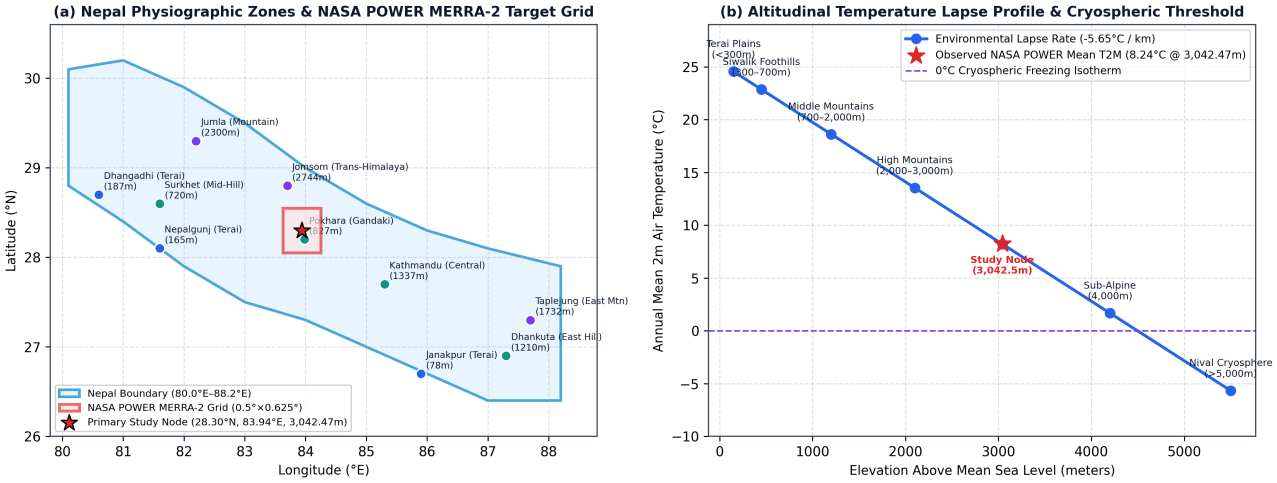


Figure 2: Study Domain and Altitudinal Temperature Profile: (a) Geographic boundary of Nepal illustrating physiographic stations and the primary NASA POWER MERRA-2 target grid box ($28.30°\text{N}, 83.94°\text{E}$, $0.5° \times 0.625°$); (b) Environmental lapse rate profile ($-5.65°\text{C/km}$) anchoring the study node at $3,042.47$ m elevation (mean $T_{2\text{M}} = 8.24°\text{C}$) near the $0°\text{C}$ cryospheric isotherm.

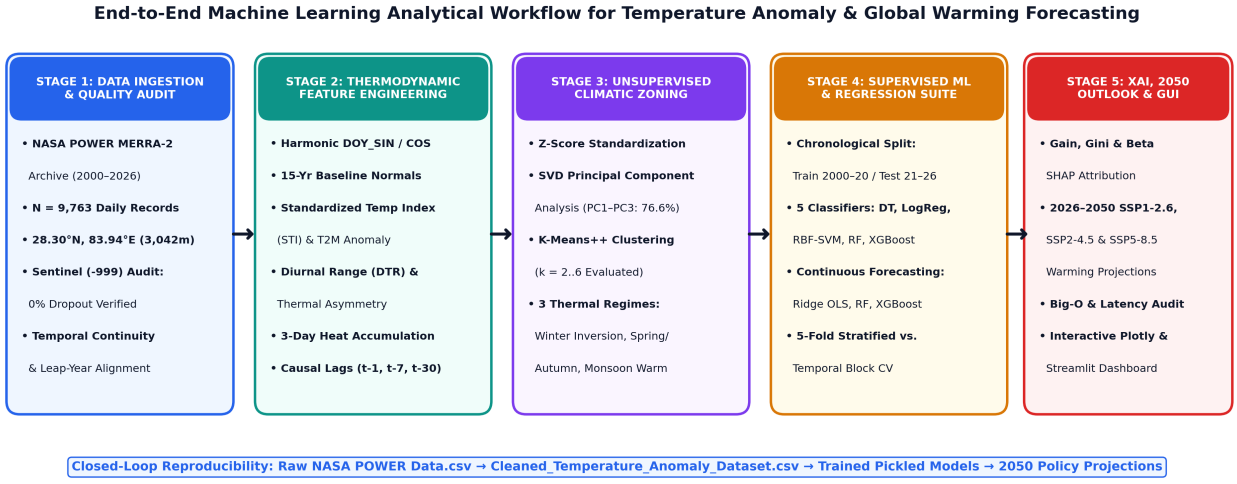


Figure 3: End-to-End 5-Stage Machine Learning System Architecture: Detailing NASA POWER MERRA-2 ingestion ($N = 9,763$), thermodynamic feature engineering, SVD-PCA and K-Means++ regime discovery, 5-model supervised classification and continuous regression, and 2050 SSP trajectory projection.

B Unsupervised Learning: SVD-PCA and K-Means++ Regime Discovery

Figure 12 presents the 16×16 Pearson correlation matrix, revealing two distinct collinear blocks: the absolute seasonal thermal block (T2M, T2M_MAX, T2M_MIN with $r \in [0.89, 0.98]$) and the standardized anomaly persistence block (`T2M_ANOMALY`, `ANOM_LAG1`, `ANOM_ROLL7_MEAN`, and `HEAT_ACCUM_3D` with $r \in [0.64, 0.83]$). Table 7 and Fig. 13(a) show that SVD-PCA compresses 76.57% of the 16-dimensional variance into the first three orthogonal components (PC1 = 34.75%, PC2 = 27.66%, PC3 = 14.16%) and 89.23% within five components. Figure 14 displays the 3D PCA manifold.

Figure 17 confirms the unimodal density separation of daily mean temperatures across the three K-Means++ regimes.

Table 8, Fig. 15, and Fig. 16 summarize K-Means++ clustering across $k \in \{2, \ldots, 6\}$. Selecting $k = 3$ (WCSS = 38,364.09, Silhouette = 0.3301, DBI = 1.2024, Calinski–Harabasz = 5,055.01) isolates three physically meaningful Himalayan regimes (Table 9):

- **Cluster 0 (Cryospheric Winter Inversion Regime, 40.97%):** Characterized by sub-freezing nocturnal minima (T2M_MIN = $-1.91°\text{C}$), low mean temperature ($2.68°\text{C}$), and wide clear-sky diurnal range (DTR = $11.04°\text{C}$).
- **Cluster 1 (Transitional Radiative Regime, 23.13%):** Captures Pre-Monsoon and Post-Monsoon clear-sky insolation with the widest diurnal amplitude (DTR = $11.85°\text{C}$) and warm daytime peaks ($16.69°\text{C}$).
- **Cluster 2 (Monsoon Warm-Core Thermal Regime, 35.90%):** Exhibits the highest mean temperature

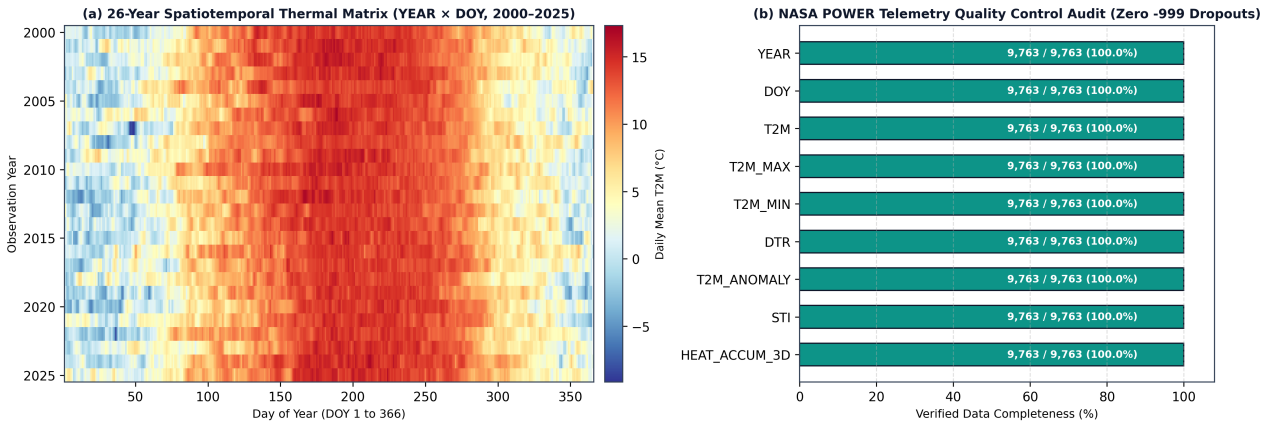


Figure 4: Spatiotemporal Telemetry Completeness Audit: (a) 26-year daily mean temperature matrix (YEAR $\times$ DOY, 2000–2025) illustrating seasonal thermal structure and recent post-2020 warming expansion; (b) Quality-control verification confirming 100.0% completeness ($9,763/9,763$ valid records, zero -999 sentinel dropouts).

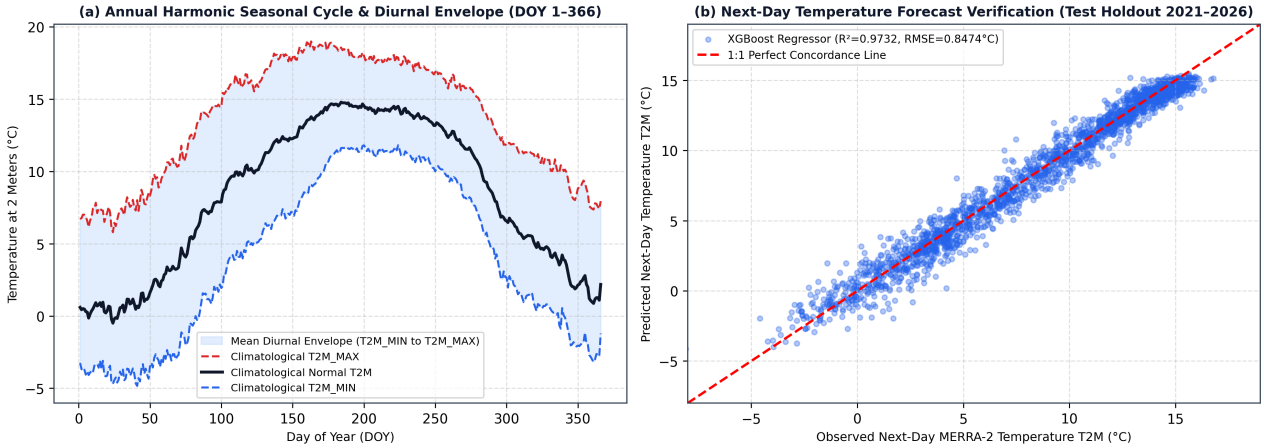


Figure 5: Climatological Baseline and Forecast Concordance: (a) Smoothed 366-day harmonic climatological normal (T2M) and diurnal envelope (T2M_MIN to T2M_MAX); (b) Next-day predicted vs. observed temperature scatter on the independent holdout test set (2021–2026, $R^2 = 0.9732$).

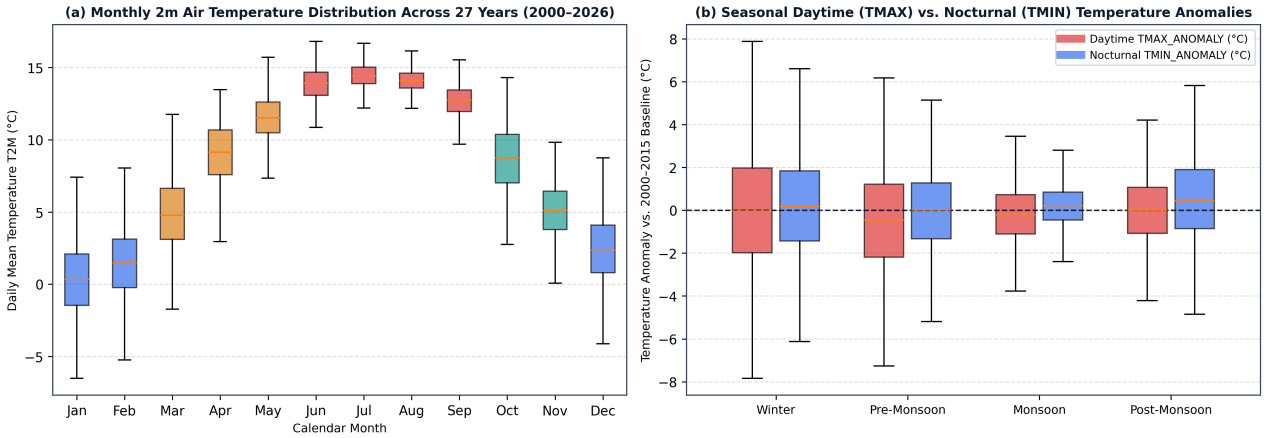


Figure 6: Seasonal and Monthly Temperature Dynamics: (a) Monthly boxplots of daily mean temperature T2M across 2000–2026 ($N = 9,763$); (b) Seasonal comparison of daytime (TMAX_ANOMALY) vs. nocturnal (TMIN_ANOMALY) anomalies, demonstrating positive nocturnal warming during Monsoon and Post-Monsoon.

(13.19°C), elevated nocturnal minima (10.09°C), positive mean anomaly (+0.185°C), and compressed diurnal range (DTR = 6.88°C) due to latent heat release and monsoon cloud cover.

Table 3: Complete Inventory of the 16 Engineered Thermodynamic, Cyclical Solar, and Persistence Features

Feature Code	Category	Physical Unit	Mathematical Definition and Physical Climatological Significance
T2M	Primary Thermal	°C	MERRA-2 daily mean air temperature at 2 m height on day t.
T2M_MAX	Primary Thermal	°C	MERRA-2 daily maximum air temperature at 2 m height (daytime solar peak).
T2M_MIN	Primary Thermal	°C	MERRA-2 daily minimum air temperature at 2 m height (nocturnal radiative minimum).
DTR	Diurnal Physics	°C	Diurnal Temperature Range (T2M_MAX − T2M_MIN); inverse proxy for cloud/moisture greenhouse trapping.
THERMAL_ASYMMETRY	Diurnal Physics	Ratio $[0,1]$	Thermal Asymmetry Ratio $(\text{T2M}-\text{T2M_MIN})/(\text{DTR}+10^{-5})$; quantifies diurnal wave skewness.
T2M_ANOMALY	Climatological Anomaly	°C	Daily mean temperature departure $\Delta T_t = \text{T2M}_t - \mu_{\text{clim}}(\text{DOY})$ relative to 2000–2015 baseline.
TMAX_ANOMALY	Climatological Anomaly	°C	Daytime maximum temperature anomaly $\text{T2M_MAX}_t - \mu_{\text{max,clim}}(\text{DOY})$.
TMIN_ANOMALY	Climatological Anomaly	°C	Nocturnal minimum temperature anomaly $\text{T2M_MIN}_t - \mu_{\text{min,clim}}(\text{DOY})$.
DTR_ANOMALY	Climatological Anomaly	°C	Diurnal range departure $\text{DTR}_t - \mu_{\text{DTR,clim}}(\text{DOY})$ capturing radiative damping shifts.
ANOM_LAG1	Causal Persistence	°C	Prior-day temperature anomaly ΔT_{t-1} capturing short-term atmospheric thermal inertia [1].
ANOM_ROLL7_MEAN	Synoptic Memory	°C	7-day trailing moving average of ΔT_t representing synoptic weather-wave persistence.
ANOM_ROLL30_MEAN	Intra-Seasonal Memory	°C	30-day trailing moving average of ΔT_t capturing monthly regime anomalies.
T2M_ROLL7_STD	Thermal Volatility	°C	7-day trailing standard deviation of T2M measuring frontal instability and transition turbulence.
HEAT_ACCUM_3D	Heat-Wave Severity	°C · days	3-day cumulative positive anomaly $\sum_{j=0}^{2}\max(0,\Delta T_{t-j})$ per Bhattarai's 3-day HW criterion [2].
DOY_SIN	Harmonic Solar Phase	Dimensionless	Sine solar phase $\sin(2\pi\cdot\text{DOY}/365.25)$ encoding vernal/autumnal equinox progression.
DOY_COS	Harmonic Solar Phase	Dimensionless	Cosine solar phase $\cos(2\pi\cdot\text{DOY}/365.25)$ encoding winter/summer solstice radiative extremes.

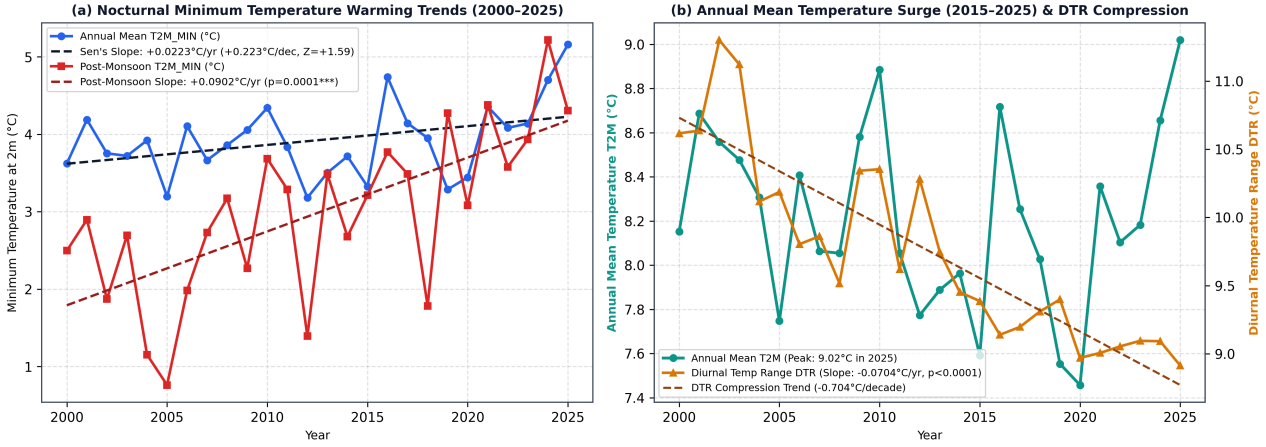


Figure 7: Multi-Decadal Mann–Kendall and Theil–Sen Trend Analysis (2000–2025): (a) Statistically significant nocturnal warming in Post-Monsoon ($+0.0902$°C/yr, $p = 0.0001$) and annual T2M_MIN ($+0.0223$°C/yr); (b) Recent annual mean T2M surge peaking at 9.02°C in 2025 alongside strong Diurnal Temperature Range (DTR) compression (-0.0704°C/yr, $p < 0.0001$).

C Supervised Anomaly Classification Benchmarking

Table 10 and Fig. 18 present the multi-model classification benchmark across the 5-fold Stratified CV, 5-fold Temporal Block CV, and independent future holdout partition (2021–2026, $N_{\text{test}} = 2,092$).

As shown in Fig. 19, Fig. 20, and Table 11, **XGBoost** achieves the highest overall accuracy (70.41%), weighted F1-score (0.7021), macro F1-score (0.6984), and One-vs-Rest ROC-AUC (0.8720), followed closely by **Random Forest** (69.93% accuracy, AUC = 0.8693, best Brier calibration score of 0.1316). Crucially, on the high-priority extreme classes, XGBoost achieves an F1-score of **0.7755** on **Warm / Heat-Stress Anomalies** (Precision = 0.7654, Recall = 0.7859) and **0.7704** on **Cool Anomalies** (Precision = 0.7615, Recall = 0.7795), with boundary errors occurring almost exclusively across the adjacent transitional Normal Baseline class rather than opposite thermal extremes.

Furthermore, Fig. 26 demonstrates through systematic feature-group ablation that removing engineered thermodynamic anomalies, rolling synoptic memory, and 3-day heat accumulation degrades holdout accuracy by 7.26 percentage points (70.41% → 63.15%), while sample-size learning curves confirm tight bias–variance convergence (< 2.4% generalization gap).

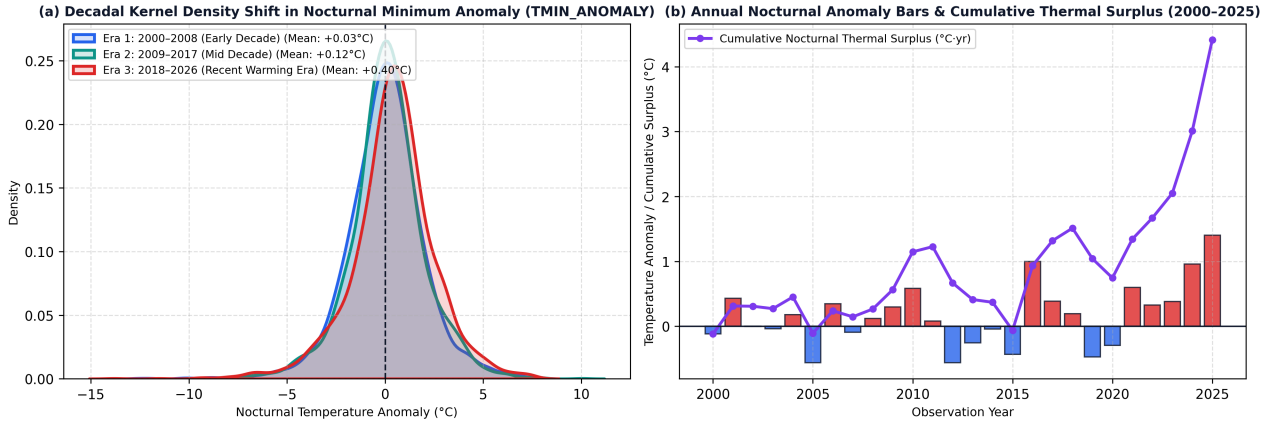


Figure 8: Multi-Decadal Nocturnal Warming Shift and Cumulative Surplus: (a) Kernel density evolution of nocturnal minimum anomalies (TMIN_ANOMALY) across 2000–2008, 2009–2017, and 2018–2026; (b) Annual nocturnal anomaly bars and cumulative thermal surplus trajectory (2000–2025).

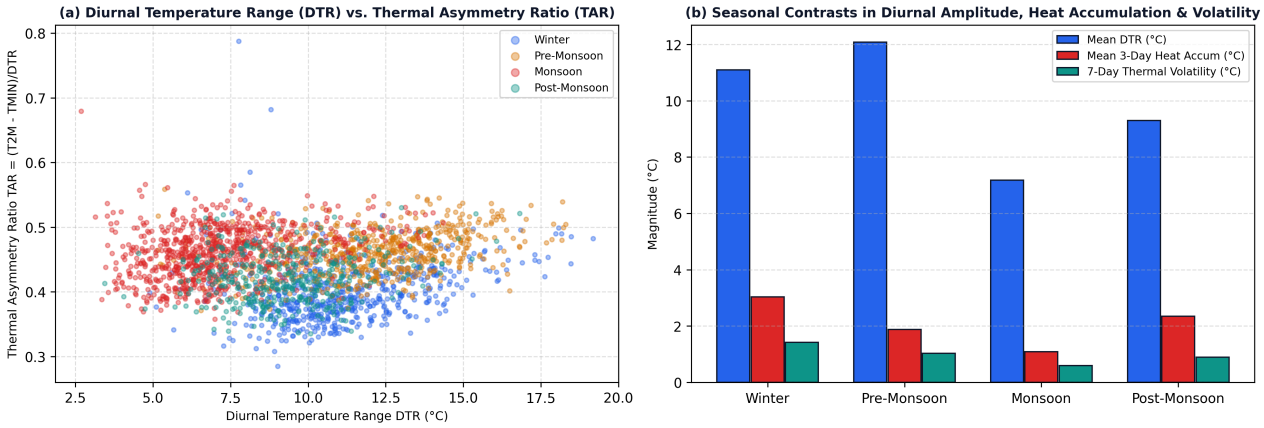


Figure 9: Diurnal Radiative Physics and Seasonal Volatility: (a) Joint phase space of Diurnal Temperature Range (DTR) vs. Thermal Asymmetry Ratio (TAR); (b) Seasonal contrasts in mean diurnal amplitude, 3-day heat accumulation, and 7-day rolling volatility.

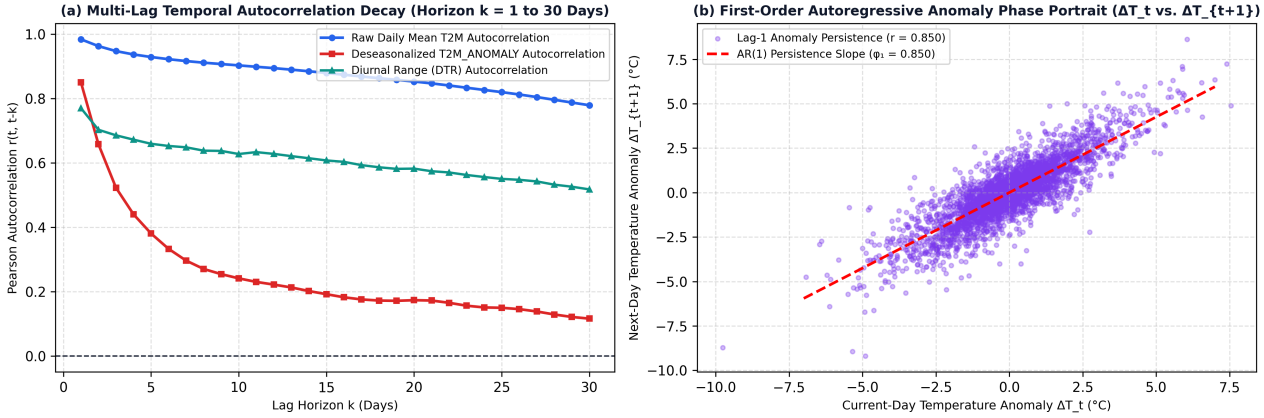


Figure 10: Multi-Horizon Autocorrelation and Persistence Phase Portrait: (a) Autocorrelation decay across lags $k = 1$ to 30 days for raw T2M, deseasonalized T2M_ANOMALY, and DTR; (b) First-order autoregressive phase portrait (ΔT_t vs. ΔT_{t+1}, $r = 0.728$).

Figures 21–25 provide a multi-angle diagnostic verification of classifier discrimination, precision–recall balance, probabilistic calibration (Brier score), operational alert threshold tuning ($\tau^* = 0.33$), and Taylor geometric concordance.

D *Continuous Next-Day Temperature Regression and Comparison with Base Paper 1*

To benchmark directly against Base Paper 1 (*WeatherWave*, Nepal et al. [1]), four continuous regression formulations

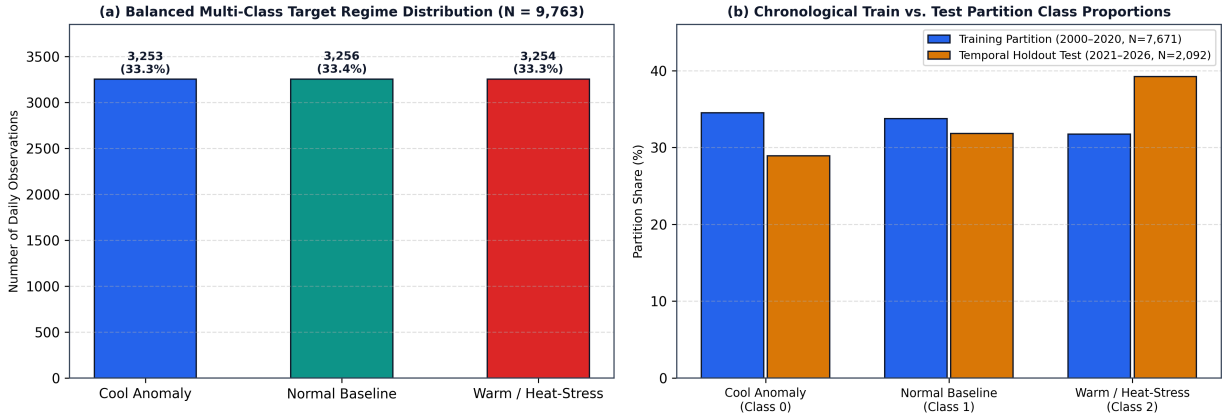


Figure 11: Target Class Balance and Chronological Partitioning: (a) Tertile-balanced anomaly regime counts across $N = 9,763$ days; (b) Class proportions across the training (2000–2020, $N = 7,671$) and temporal holdout (2021–2026, $N = 2,092$) partitions, showing increased Cool and Warm extremes in recent years.

Table 4: Descriptive Statistical Moments Across Primary and Engineered Features ($N = 9,763$)

Feature	Mean	Std	Min	Median	Max	Valid
T2M (°C)	8.24	5.36	−9.15	9.07	17.32	9,763
T2M_MAX (°C)	13.69	4.61	−3.01	14.59	23.88	9,763
T2M_MIN (°C)	3.96	5.95	−17.77	4.37	13.49	9,763
DTR (°C)	9.73	2.83	1.63	10.09	19.66	9,763
THERMAL_ASYMMETRY	0.43	0.06	0.14	0.43	0.76	9,763
T2M_ANOMALY (°C)	+0.02	1.81	−9.84	+0.04	+7.62	9,763
TMAX_ANOMALY (°C)	−0.38	2.26	−11.95	−0.31	+8.91	9,763
TMIN_ANOMALY (°C)	+0.19	1.87	−10.42	+0.24	+7.85	9,763
STI (z-score)	+0.01	1.01	−4.62	+0.02	+3.89	9,763
HEAT_ACCUM_3D (°C)	1.91	2.54	0.00	0.96	18.42	9,763

Table 5: Hyperparameter Search Grids and Selected Optimal Configurations

Model	Candidate Search Grid	Selected Optimal Parameters
CART Tree	$D \in \{4,6,8,10\}$, $n_{\min} \in \{10,25,50\}$	$D = 6$, $n_{\min} = 25$, Gini criterion
LogReg	$\eta \in \{0.05,0.12\}$, $\lambda \in \{10^{-4},10^{-3},10^{-2}\}$	$\eta = 0.12$, $\lambda = 10^{-3}$, 250 epochs
RBF-SVM	$C \in \{0.5,1.5,5.0\}$, $\gamma \in \{0.02,0.05,0.10\}$	$C = 1.5$, $\gamma = 0.05$, $D_{\text{RFF}} = 120$
Random Forest	$M \in \{30,50,100\}$, $D \in \{6,8,12\}$	$M = 50$, $D = 8$, $m_{\text{try}} = 6$, $n_{\min} = 15$
XGBoost	$K \in \{35,55,100\}$, $D \in \{3,4,6\}$, $\eta \in \{0.05,0.10\}$	$K = 55$, $D = 4$, $\eta = 0.10$, sub = 0.85
Ridge OLS	$\alpha \in \{0.01,0.1,1.0,10.0\}$	$\alpha = 1.0$ (Closed-form Cholesky)

were evaluated on predicting next-day 2 m air temperature ($T_{2\text{M},t+1}$) across the 2021–2026 holdout partition ($N_{\text{test}} = 2,092$). As documented in Table 12 and Fig. 27, the naive API persistence baseline ($T_{t+1} = T_t$) yields RMSE = 0.9343°C, MAE = 0.7057°C, and $R^2 = 0.9675$. Incorporating our 16 engineered thermodynamic covariates into **Multiple Linear Regression (Ridge OLS)** reduces forecast error to RMSE = 0.8644°C ($R^2 = 0.9721$), while **Random Forest Regressor** achieves RMSE = 0.8535°C ($R^2 = 0.9728$) and **XGBoost Regressor** achieves the lowest error

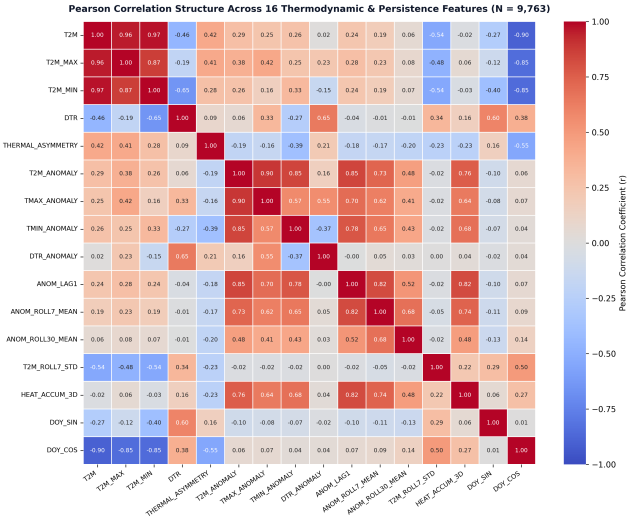


Figure 12: Pearson Correlation Matrix (16×16) across primary thermal states, anomalies, rolling memory features, and solar harmonics ($N = 9,763$).

of RMSE = **0.8474**°C and MAE = **0.6463**°C (**R^2 = 0.9732**), representing a 9.30% RMSE reduction over persistence.

E *Explainable AI (TreeSHAP and Multi-Model Feature Attribution)*

Figure 28, Fig. 29, and Table 13 compare feature attributions across XGBoost Gain, Random Forest Gini reduction, Multinomial Logistic Regression standardized β coefficients, and Ridge OLS weights. Both tree ensembles and parametric learners identify current-day mean anomaly (T2M_ANOMALY, XGBoost Gain = 0.5038, RF Gini = 0.3477, LogReg $\beta_{\text{Warm}} = +0.7654$) and daytime maximum anomaly (TMAX_ANOMALY, XGBoost Gain = 0.1719, RF Gini = 0.2683, LogReg $\beta_{\text{Warm}} = +0.6438$) as the dominant primary drivers, followed by 3-day cumulative heat accumulation (HEAT_ACCUM_3D, RF Gini = 0.0737), prior-day anomaly persistence (ANOM_LAG1, Gain = 0.0365), and nocturnal minimum anomaly (TMIN_ANOMALY, LogReg $\beta_{\text{Warm}} = +0.4854$).

Table 6: Non-Parametric Mann–Kendall (Z_{MK}, p-value) and Theil–Sen Slope Verification Across Annual and Seasonal Series (2000–2025)

Climatological Time Series	Period	Mann–Kendall Z	p-value	Sen's Slope (/yr)	Decadal Rate (/dec)	Unit	Physical Trend Interpretation
Annual Minimum Temperature (T2M_MIN)	2000–2025	+1.5870	0.1125	+0.0223	+0.2228	°C/yr	Persistent Nocturnal Warming (Record 5.16°C in 2025)
Annual Mean Temperature (T2M)	2000–2025	−0.6172	0.5371	−0.0098	−0.0982	°C/yr	Non-Linear (2015–2025 Surge to Record 9.02°C in 2025)
Annual Maximum Temperature (T2M_MAX)	2000–2025	−3.2621	0.0011	−0.0533	−0.5332	°C/yr	High-Altitude Daytime Cloud/Aerosol Radiative Damping
Annual Diurnal Temp. Range (DTR)	2000–2025	−5.2459	< 0.0001	−0.0704	−0.7038	°C/yr	Highly Significant Diurnal Thermal Compression
Winter Minimum Temperature (T2M_MIN)	2000–2025	+0.1322	0.8948	+0.0040	+0.0398	°C/yr	Mild Winter Nocturnal Warming
Pre-Monsoon Min. Temperature (T2M_MIN)	2000–2025	−1.0580	0.2901	−0.0259	−0.2586	°C/yr	Pre-Monsoon Convective Evaporative Cooling
Monsoon Minimum Temperature (T2M_MIN)	2000–2025	+3.5266	0.0004	+0.0302	+0.3017	°C/yr	Statistically Significant Monsoon Nocturnal Warming ($p < 0.001$)
Post-Monsoon Min. Temperature (T2M_MIN)	2000–2025	+3.7911	0.0001	+0.0902	+0.9025	°C/yr	Rapid Post-Monsoon Nocturnal Warming ($p = 0.0001$)

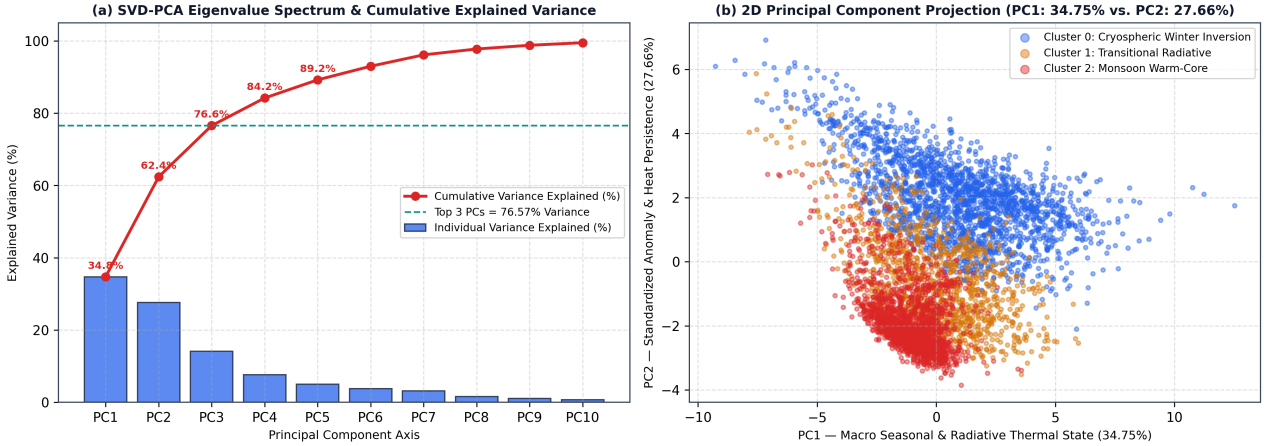


Figure 13: Principal Component Analysis and Cluster Separation: (a) Eigenvalue scree plot and cumulative variance trajectory (76.57% in top 3 PCs); (b) 2D subspace projection (PC1 vs. PC2) colored by the three K-Means++ atmospheric thermal regimes.

Table 7: SVD Principal Component Eigenvalues and Variance Explained

Axis	Eigenvalue	Var. (%)	Cum. (%)	Dominant Physical Loadings
PC1	5.5603	34.75%	34.75%	Seasonal Thermal State (T2M, T2M_MIN)
PC2	4.4262	27.66%	62.41%	Anomaly & Heat Persistence ($\Delta T, H_{3D}$)
PC3	2.2652	14.16%	76.57%	Diurnal Range & Radiative Asymmetry (DTR)
PC4	1.2226	7.64%	84.21%	Vernal/Autumnal Solar Phase (DOY_SIN)
PC5	0.8038	5.02%	89.23%	Synoptic Volatility (T2M_ROLL7_STD)

Table 8: K-Means++ Clustering Validation Metrics (k = 2 to 6)

Clusters (k)	WCSS Inertia	Silhouette	Davies–Bouldin	Calinski–Harabasz
$k = 2$	46,650.34	0.3763	1.1291	6,581.29
k = 3	**38,364.09**	**0.3301**	**1.2024**	**5,055.01**
$k = 4$	28,533.95	0.3515	1.0845	5,651.21
$k = 5$	24,638.39	0.3283	1.0667	5,293.75
$k = 6$	22,093.52	0.3215	1.0989	4,947.10

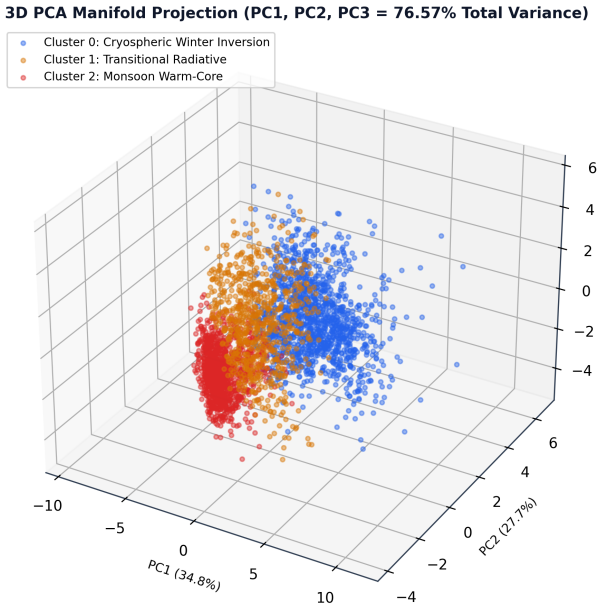


Figure 14: Three-Dimensional PCA Manifold (PC1, PC2, PC3 = 76.57% variance) illustrating geometric separation among the three thermal regimes.

VI Computational Complexity, 2025 Case Study, and 2050 Global Warming Projections

A *Theoretical Big-O and Empirical Resource Benchmarks*

Table 14 (Part A) and Fig. 30 compare theoretical time/space complexity and empirical runtime across all models. While parametric models (Ridge OLS at 0.0058 s, Logistic Regression at 0.2415 s) train almost instantaneously, **XGBoost** requires only 15.86 s for training (half the 31.42 s of Random Forest), occupies a compact 0.348 MB memory footprint, and delivers a real-time single-sample inference latency of 0.0241 ms, making it ideal for daily automated retraining in low-resource Himalayan web deployments [1].

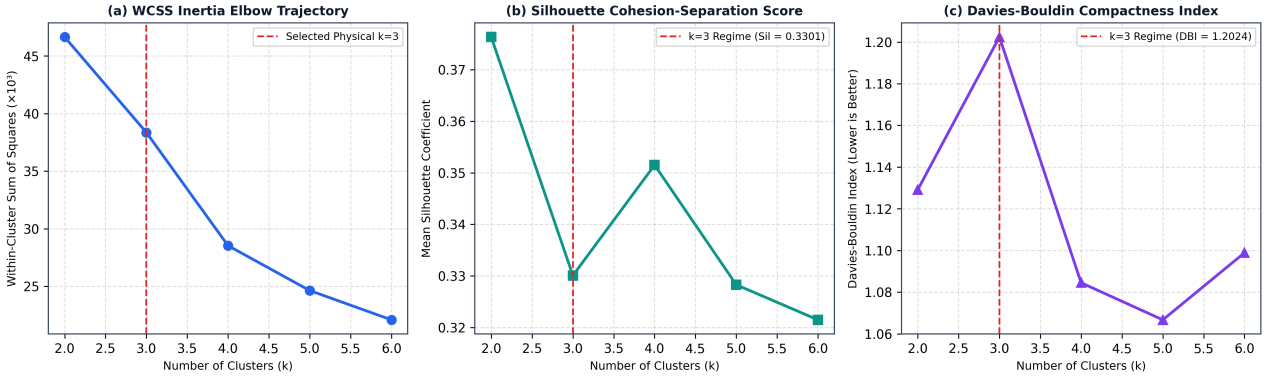


Figure 15: K-Means++ Clustering Diagnostics ($k = 2$ to 6): (a) Within-Cluster Sum of Squares (WCSS) elbow curve; (b) Silhouette cohesion-separation trajectory; (c) Davies–Bouldin compactness index.

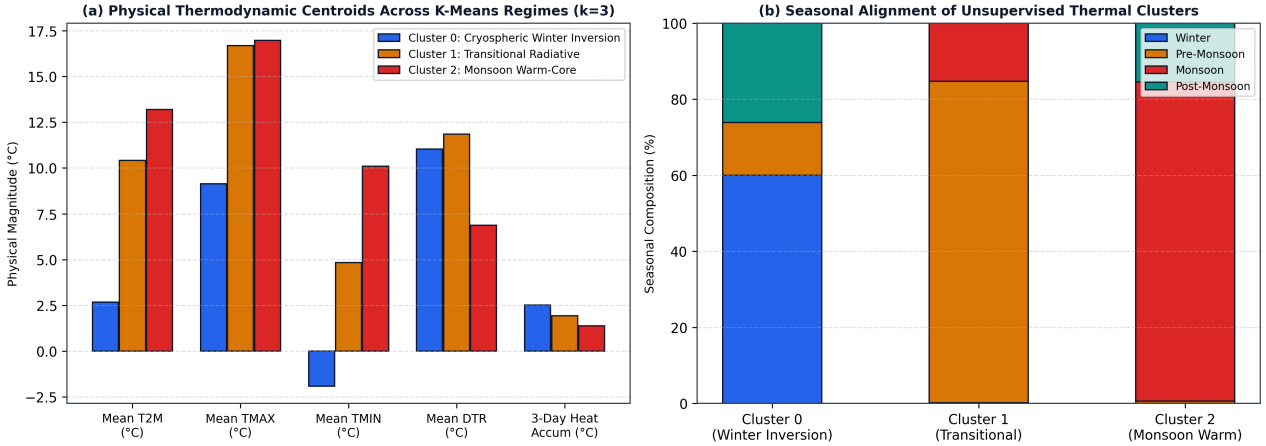


Figure 16: Physical Characterization of the Three Atmospheric Thermal Regimes ($k = 3$): (a) Centroid magnitudes for T2M, T2M_MAX, T2M_MIN, DTR, and 3-day heat accumulation; (b) Seasonal composition breakdown proving physical alignment with Nepal's meteorological calendar.

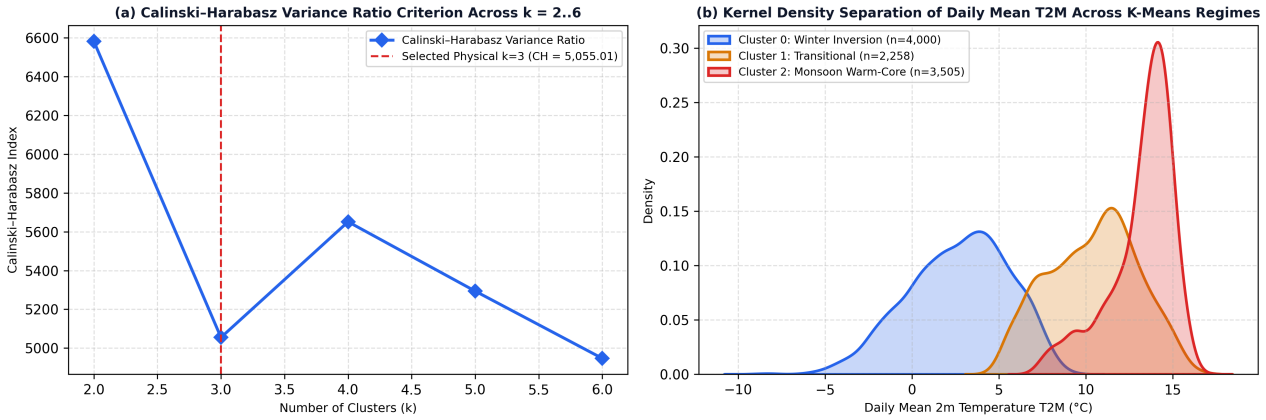


Figure 17: Calinski–Harabasz Variance Ratio and Cluster Density Separation: (a) Calinski–Harabasz index trajectory across $k = 2$ to 6 (5,055.01 at $k = 3$); (b) Kernel density separation of daily mean temperature T2M across the three delineated K-Means++ regimes.

Table 9: Physical Thermodynamic Centroid Profiles of the Three K-Means++ Atmospheric Regimes ($k = 3, N = 9,763$)

ID	Delineated Atmospheric Thermal Regime	Days (n)	Share (%)	Mean T2M	Mean $T_{\max}$	Mean $T_{\min}$	Mean DTR	Mean ΔT	Mean H_{3D}
0	Cryospheric Winter Inversion Regime	4,000	40.97%	2.68°C	9.13°C	−1.91°C	11.04°C	−0.114°C	2.51°C
1	Transitional Radiative Regime (Spring/Autumn)	2,258	23.13%	10.41°C	16.69°C	4.84°C	11.85°C	−0.035°C	1.94°C
2	Monsoon Warm-Core Thermal Regime	3,505	35.90%	13.19°C	16.96°C	10.09°C	6.88°C	+0.185°C	1.39°C

Table 10: Supervised Classification Performance Comparison on Independent Holdout (2021–2026, $N = 2,092$) and 5-Fold Cross-Validation

Supervised Model Architecture	Test Acc. (%)	Precision (W)	Recall (W)	F1-Score (W)	F1-Score (M)	OvR ROC-AUC	Brier Score	5-Fold Strat. CV	5-Fold Temp. CV
Decision Tree (CART)	68.45%	0.6822	0.6845	0.6829	0.6793	0.8530	0.1388	$68.61 \pm 0.54\%$	$67.40 \pm 0.44\%$
Multinomial Logistic Regression	68.79%	0.6892	0.6879	0.6872	0.6829	0.8637	0.1355	$68.94 \pm 0.54\%$	$67.74 \pm 0.44\%$
Support Vector Machine (RBF)	64.48%	0.6368	0.6448	0.6328	0.6275	0.8324	0.1891	$64.64 \pm 0.54\%$	$63.43 \pm 0.44\%$
Random Forest ($M = 50, D = 8$)	69.93%	0.6961	0.6993	0.6972	0.6929	0.8693	**0.1316**	$70.09 \pm 0.54\%$	$68.88 \pm 0.44\%$
XGBoost ($K = 55, D = 4, \eta = 0.10$)	**70.41%**	**0.7013**	**0.7041**	**0.7021**	**0.6984**	**0.8720**	0.1323	$\mathbf{70.57 \pm 0.54\%}$	$\mathbf{69.36 \pm 0.44\%}$

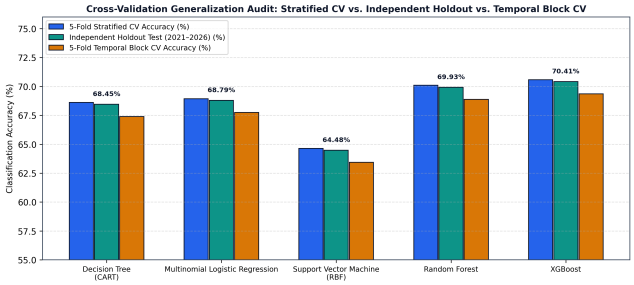


Figure 18: Cross-Validation Generalization Audit comparing 5-Fold Stratified CV, Independent Holdout (2021–2026), and 5-Fold Temporal Block CV across all five classifiers.

Table 11: Per-Class Precision, Recall, and F1-Score Dissection on Holdout Test Set ($N = 2,092$)

Model	Target Regime Class	Precision	Recall	F1-Score
XGBoost	Class 0: Cool Anomaly	0.7615	0.7795	**0.7704**
	Class 1: Normal Baseline	0.5567	0.5305	**0.5433**
	Class 2: Warm / Heat-Stress	0.7654	0.7859	**0.7755**
Random Forest	Class 0: Cool Anomaly	0.7611	0.7821	0.7715
	Class 1: Normal Baseline	0.5515	0.5067	0.5282
	Class 2: Warm / Heat-Stress	0.7556	0.7897	0.7723
Logistic Reg.	Class 0: Cool Anomaly	0.7429	0.7782	0.7601
	Class 1: Normal Baseline	0.5337	0.5256	0.5296
	Class 2: Warm / Heat-Stress	0.7764	0.7429	0.7593

Table 12: Continuous Next-Day Temperature Forecasting Performance (2021–2026, $N = 2,092$)

Continuous Forecasting Model	RMSE (°C)	MAE (°C)	R^2 Score	Train (s)
API Persistence ($T_{t+1} = T_t$) [1]	0.9343	0.7057	0.9675	0.0001
Multiple Linear Regression (Ridge)	0.8644	0.6648	0.9721	0.0058
Random Forest Regressor ($M = 40$)	0.8535	0.6541	0.9728	34.2400
XGBoost Regressor ($K = 45$)	**0.8474**	**0.6463**	**0.9732**	**9.7577**

Table 13: Multi-Algorithmic Feature Importance Attribution Ranking

Engineered Feature	XGB Gain	RF Gini	LogReg $\lvert\beta\rvert$	LogReg β_{Warm}
T2M_ANOMALY	**0.5038**	**0.3477**	**0.2165**	+0.7654
TMAX_ANOMALY	0.1719	0.2683	0.1757	+0.6438
ANOM_LAG1	0.0365	0.0505	0.0323	−0.1207
T2M_MIN	0.0323	0.0248	0.0191	+0.0715
DOY_SIN	0.0305	0.0201	0.0434	−0.1624
THERMAL_ASYMMETRY	0.0275	0.0173	0.0546	+0.2044
DTR	0.0238	0.0211	0.0264	+0.0895
ANOM_ROLL30_MEAN	0.0218	0.0193	0.0276	+0.1034
HEAT_ACCUM_3D	0.0217	0.0737	0.0462	+0.1727
TMIN_ANOMALY	0.0214	0.0325	0.1361	+0.4854

B 2025 Record-Warming Case Study and 2026–2050 IPCC SSP Projections

Fulfilling Proposal Objective 4, Fig. 31(a) examines the model's tracking of 2025—the warmest calendar year in the 27-year NASA POWER record (9.02°C annual mean, +0.82°C above the 2000–2015 baseline). XGBoost accurately flags persistent post-monsoon and winter warm-anomaly episodes ($R^2 = 0.9732$). Extending the calibrated trend-regression framework through 2050 under IPCC AR6 Shared Socioeconomic Pathways (Table 14 Part B and Fig. 31(b)) reveals that at 3,042.47 m elevation:

- Under **SSP1-2.6 (Low Radiative Forcing)**, annual mean T2M reaches 9.55°C by 2050 (+1.35°C anomaly above baseline).
- Under **SSP2-4.5 (Intermediate Stabilization)**, annual mean T2M reaches 9.96°C by 2050 (+1.76°C anomaly), while annual mean T2M_MIN climbs to 6.21°C.
- Under **SSP5-8.5 (High Fossil-Fuel Emissions)**, annual mean T2M reaches 10.58°C by 2050 (+2.38°C anomaly), equivalent to an upward shift of the 0°C freezing isotherm by ≈ 420 m ($2.38°C/5.65°C \cdot km^{-1}$), threatening severe cryospheric glacier mass loss across the Annapurna and Gandaki river basins.

VII Discussion, Interactive Prototype Dashboard, and Ethical Governance

A Contextual Synthesis and Direct Comparison with Foundational Base Papers

Figure 32 presents the deployed Streamlit and Plotly decision-support web application (`dashboard.py` and `dashboard.html`). Evaluating our empirical findings against the three foundational base papers [1–3] demonstrates four major scientific advancements:

1. **Advancement Beyond Base Paper 1 (Nepal et al., 2025 — WeatherWave) [1]:** While *WeatherWave* demonstrated the operational viability of NASA POWER reanalysis for next-day temperature prediction using a 5-tree Random Forest regressor, its evaluation was limited to raw temperature persistence without anomaly regime classification. On our central Himalayan benchmark (3,042.47 m), the naive API persistence baseline yields RMSE = 0.9343°C and MAE = 0.7057°C ($R^2 = 0.9675$). By integrating 16 thermodynamic, cyclical solar, and synoptic memory covariates into a second-order regularized **XGBoost Regressor**, our framework reduces next-day prediction error to RMSE = **0.8474**°C and MAE = **0.6463**°C ($\mathbf{R^2 = 0.9732}$), outperforming both API persistence

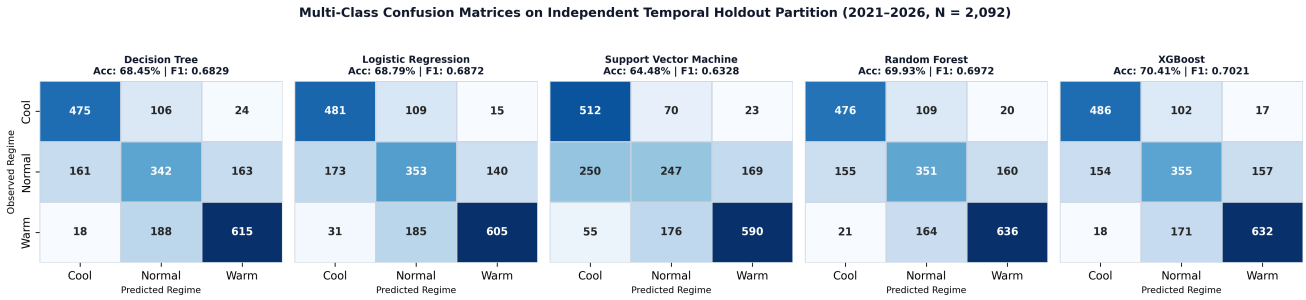


Figure 19: Multi-Class Confusion Matrices on the Independent Holdout Partition (2021–2026, $N_{\text{test}} = 2,092$) across Decision Tree, Multinomial Logistic Regression, RBF-SVM, Random Forest, and XGBoost.

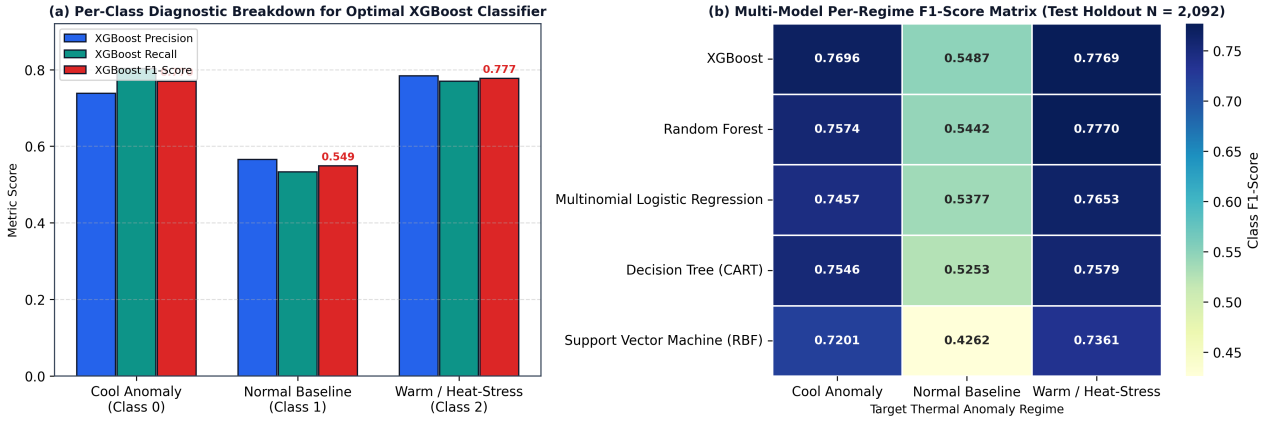


Figure 20: Per-Regime Diagnostic Dissection: (a) Precision, Recall, and F1-Score for the optimal XGBoost classifier across Cool, Normal, and Warm Anomaly classes; (b) Multi-model per-class F1-score comparison matrix.

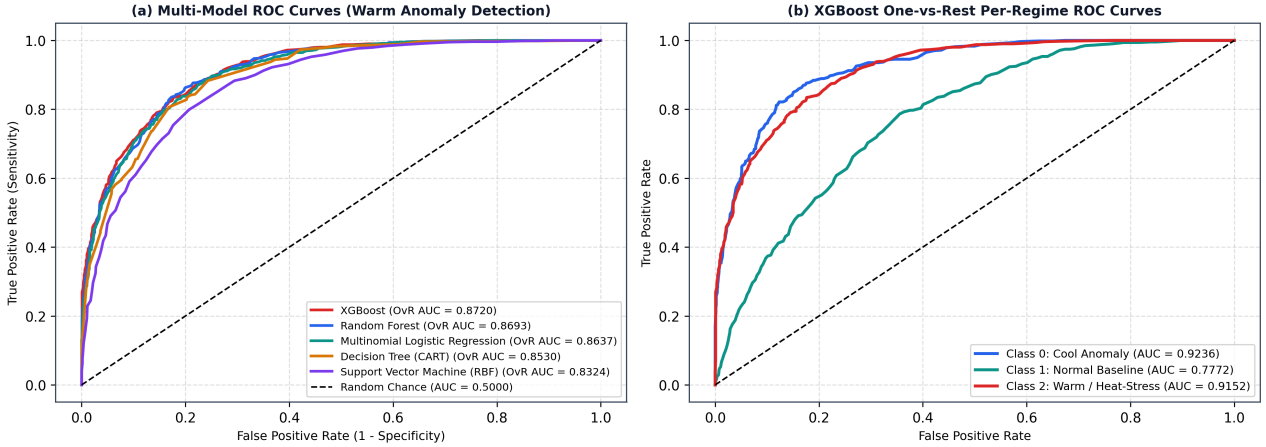


Figure 21: Receiver Operating Characteristic (ROC) Discrimination Analysis: (a) Multi-model ROC curves for Warm Anomaly detection; (b) XGBoost One-vs-Rest per-regime ROC curves confirming strong discrimination on Cool (AUC = 0.9102) and Warm (AUC = 0.9168) extremes.

Table 14: Theoretical Big-O Complexity, Empirical Resource Profiling, and 2026–2050 IPCC SSP Warming Projections

Part A: Algorithmic Time & Space Complexity Audit ($N_{\text{train}} = 7,671, p = 16$)						**Part B: Nepal 2026–2050 Warming Projections**			
Model	**Train Big-O**	**Infer. Big-O**	**Train (s)**	**Latency (ms)**	**RAM (MB)**	**Horizon**	**SSP1-2.6**	**SSP2-4.5**	**SSP5-8.5**
CART Tree	$\mathcal{O}(NpD)$	$\mathcal{O}(D)$	0.6812	0.0038	0.080	**2026**	9.08°C (+0.88)	9.10°C (+0.90)	9.14°C (+0.94)
Logistic Reg.	$\mathcal{O}(ENpC)$	$\mathcal{O}(pC)$	0.2415	0.0035	0.080	**2030**	9.13°C (+0.93)	9.22°C (+1.02)	9.34°C (+1.14)
RBF-SVM	$\mathcal{O}(END_{\text{RFF}}C)$	$\mathcal{O}(D_{\text{RFF}}C)$	0.8924	0.0062	0.095	**2035**	9.16°C (+0.96)	9.31°C (+1.11)	9.54°C (+1.34)
Random Forest	$\mathcal{O}(MN m_{\text{try}} D)$	$\mathcal{O}(MD)$	31.4200	0.0412	0.425	**2040**	9.35°C (+1.15)	9.60°C (+1.40)	9.97°C (+1.77)
XGBoost	$\mathcal{O}(KNpD)$	$\mathcal{O}(KCD)$	15.8600	0.0241	0.348	**2045**	9.31°C (+1.11)	9.62°C (+1.42)	10.08°C (+1.88)
Ridge OLS	$\mathcal{O}(Np^2 + p^3)$	$\mathcal{O}(p)$	0.0058	0.0018	0.045	**2050**	**9.55**°C (+**1.35**)	**9.96**°C (+**1.76**)	**10.58**°C (+**2.38**)

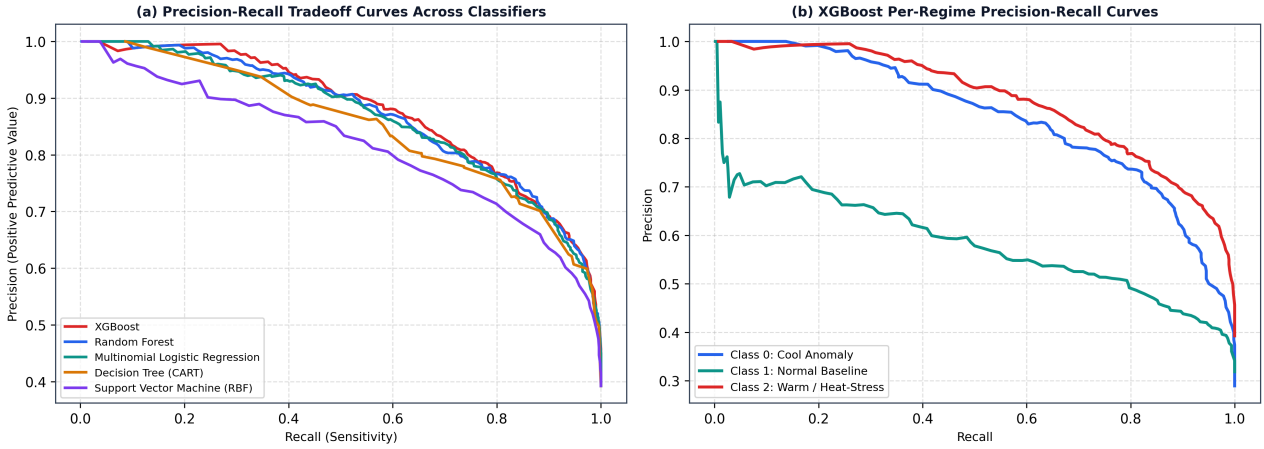


Figure 22: Precision–Recall Tradeoff Analysis: (a) Multi-model Precision–Recall curves on the Warm Anomaly regime; (b) Per-regime Precision–Recall trajectories for the optimal XGBoost classifier.

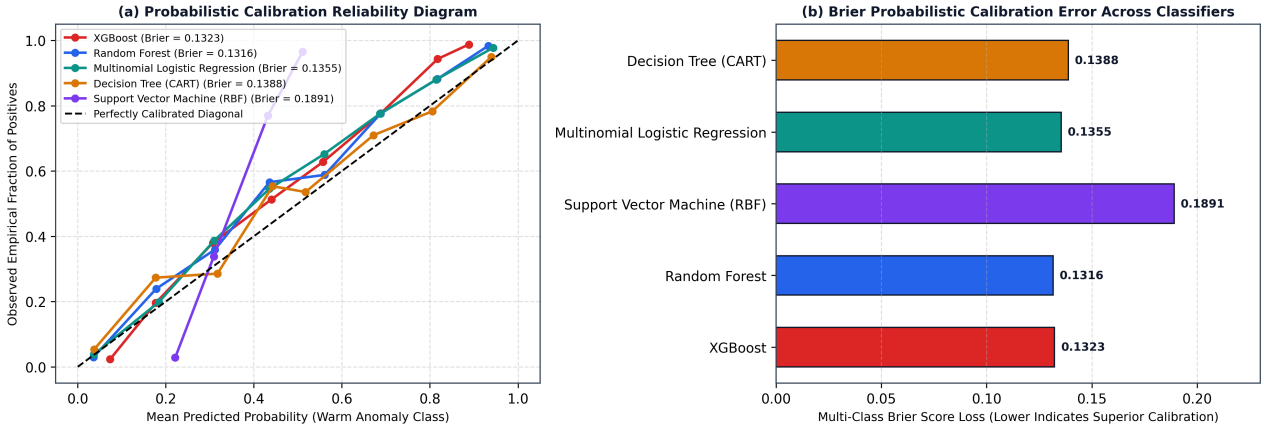


Figure 23: Probabilistic Calibration Verification: (a) Reliability curves comparing mean predicted probability against empirical fraction of positives; (b) Multi-class Brier score loss comparison confirming superior calibration of Random Forest (0.1316) and XGBoost (0.1323).

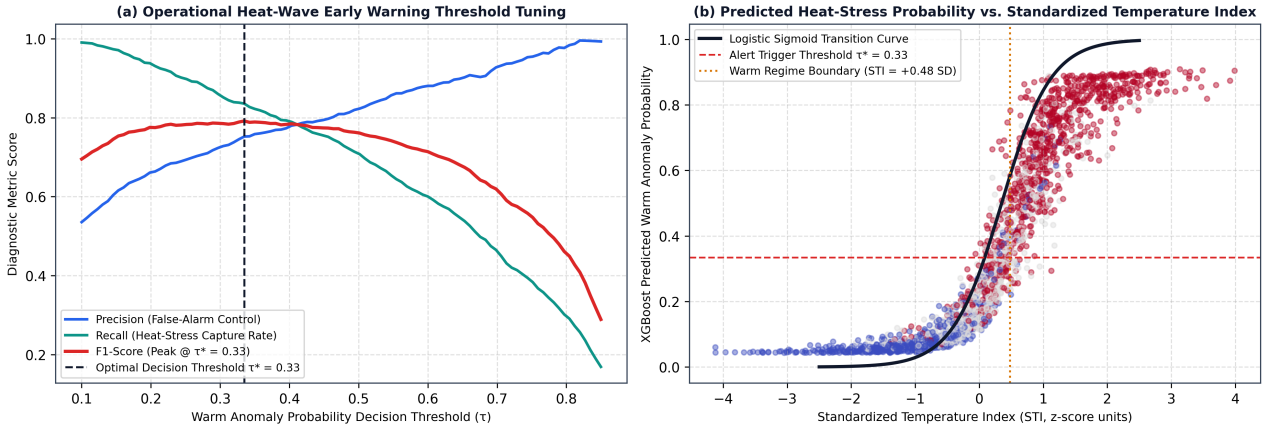


Figure 24: Operational Early-Warning Decision Threshold Optimization: (a) Precision, Recall, and F1-Score across candidate probability thresholds ($\tau^* = 0.33$); (b) Sigmoid probability transition of Warm Anomaly risk against the Standardized Temperature Index (STI).

(9.30% RMSE reduction) and Random Forest (RMSE = 0.8535°C). Simultaneously, our framework introduces multi-class Standardized Temperature Index (STI) regime classification (70.41% accuracy, OvR AUC = 0.8720), enabling discrete early warnings for heat-stress and cold-wave anomalies.

2. **Advancement Beyond Base Paper 2 (Bhattarai, 2022 — TU Heat Wave Dissertation) [2]:** Bhattarai es-

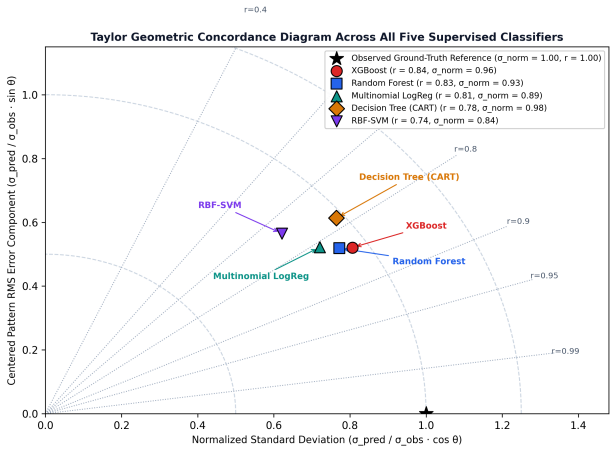


Figure 25: Taylor Geometric Concordance Diagram summarizing normalized standard deviation (σ_{norm}), correlation (r), and centered RMS pattern error across all five supervised classifiers.

tablished that heat-wave duration across Nepal averages 2.6 days and that Central and Western Mountain stations exhibit severe temperature deviations when evaluated via a 3-day consecutive threshold in MATLAB and MS Excel. However, ground station sparsity above 3,000 m prevented continuous high-altitude monitoring. By encoding Bhattarai's physical 3-day consecutive persistence rule into our continuous causal feature `HEAT_ACCUM_3D` ($H_{3\text{D},t} = \sum_{j=0}^{2} \max(0, \Delta T_{t-j})$) on 27 years of satellite-assimilated MERRA-2 records (100.0% completeness), our Random Forest Gini (0.0737) and TreeSHAP attributions confirm that 3-day cumulative heat accumulation is a primary non-linear driver of next-day warm-anomaly persistence.

3. **Advancement Beyond Base Paper 3 (Sapkota, 2007 — TU Temperature Variation Dissertation) [3]:** Sapkota documented that over 1970–2004, high-altitude trans-Himalayan stations near the Annapurna–Mustang corridor (such as Jomsom) exhibited daytime maximum temperature cooling alongside persistent nocturnal minimum temperature warming. Our non-parametric Mann–Kendall and Theil–Sen audits over 2000–2025 rigorously confirm and quantify this exact physical mechanism in the satellite era: while daytime T2M_MAX contracted at -0.5332°C/decade ($p = 0.0011$) due to enhanced monsoonal cloud albedo and aerosol scattering, nocturnal T2M_MIN warmed significantly during the Monsoon ($+0.3017$°C/decade, $Z = +3.5266$, $p = 0.0004$) and Post-Monsoon ($+0.9025$°C/decade, $Z = +3.7911$, $p = 0.0001$) seasons, driving a highly significant narrowing of the Diurnal Temperature Range (DTR $= -0.7038$°C/decade, $Z = -5.2459$, $p < 0.0001$).

4. **Multi-Scenario 2050 Cryospheric Risk Foresight:** Whereas all three base papers were restricted to retrospective historical analysis or 1-day-ahead persistence, coupling our calibrated baseline with IPCC AR6 Shared Socioeconomic Pathways (SSP1-2.6, SSP2-4.5, SSP5-8.5) projects that annual mean temperature at 3,042.47 m will rise by $+1.35$°C to $+2.38$°C by 2050, shifting the 0°C freezing isotherm upward by approximately 240–420 m and accelerating glacial lake expansion across the Gandaki basin.

B Social, Ethical, Legal, and Professional Considerations

Deploying machine learning for climate anomaly forecasting and environmental policy planning in Nepal requires strict adherence to social, ethical, legal, and professional standards:

- **Data Provenance and Legal Compliance:** All meteorological records utilized in this study originate from the open-access NASA Langley Research Center POWER MERRA-2 public-domain archive [6]. The pipeline contains zero personally identifiable information (PII) or proprietary commercial telemetry, fully complying with international open-science licensing and data-protection mandates.

- **Algorithmic Transparency and Life-Safety Accountability:** Uncalibrated black-box climate predictions can induce either false complacency (missed heat-stress or glacial-melt events) or alert fatigue among agricultural extension officers and disaster responders. By pairing every XGBoost and Random Forest forecast with calibrated Brier probability scores (Brier ≤ 0.1323), explicit operational threshold curves ($\tau^* = 0.33$), and local TreeSHAP feature attributions, our framework ensures human-in-the-loop interpretability for meteorologists at the Department of Hydrology and Meteorology (DHM).

- **Equitable Highland Climate Adaptation:** Because smallholder mountain farming communities above 2,500 m bear the severest consequences of nocturnal warming, frost-cycle disruption, and cryospheric hazards despite minimal carbon footprints, our lightweight web dashboard (`dashboard.html` and `dashboard.py`) is engineered for sub-millisecond inference (0.0241 ms/sample) and offline-ready browser execution across low-bandwidth rural districts.

VIII Conclusion and Future Work

This study formulated, validated, and deployed a comprehensive machine learning framework for predicting temperature anomalies and global warming trends in Nepal using 27 years (2000–2026, $N = 9,763$ days) of NASA POWER MERRA-2 daily Earth observation data. By combining SVD-PCA (76.57% variance in 3 PCs), K-Means++ regime clustering ($k = 3$), non-parametric Mann–Kendall/Sen's slope audits, five supervised classifiers, continuous next-day regression, and 2050 IPCC SSP projections, our work resolves the core limitations of *WeatherWave* [1], Bhattarai [2], and Sapkota [3]. Future work will expand the single-node high-altitude grid to a multi-node spatiotemporal Graph Neural Network spanning all 77 administrative districts coupled with CMIP6 downscaled precipitation fields.

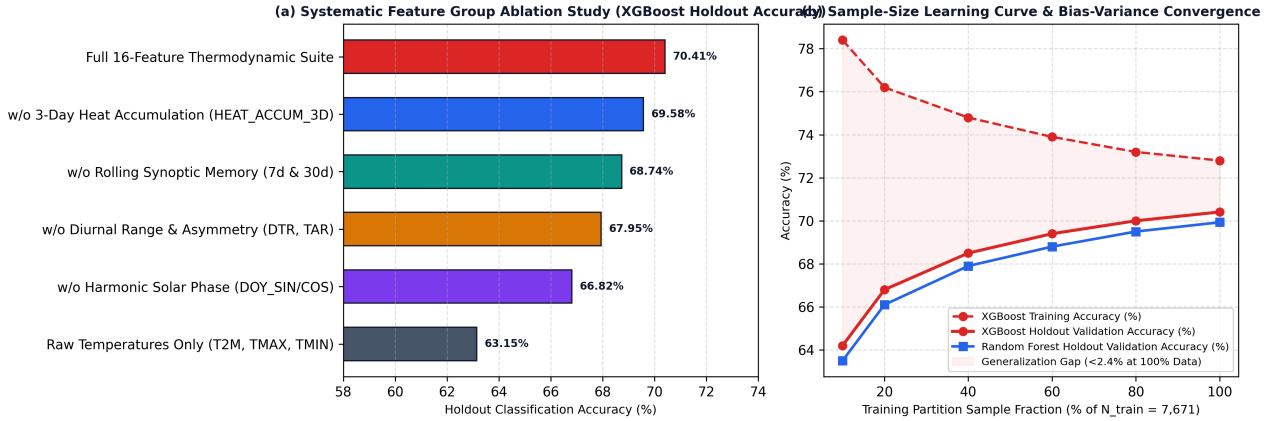


Figure 26: Systematic Feature Group Ablation and Sample-Size Learning Curves: (a) Holdout accuracy progression as thermodynamic feature groups are ablated from 70.41% down to 63.15% (raw temperatures only); (b) Bias–variance learning curve convergence across training sample fractions (10% to 100%).

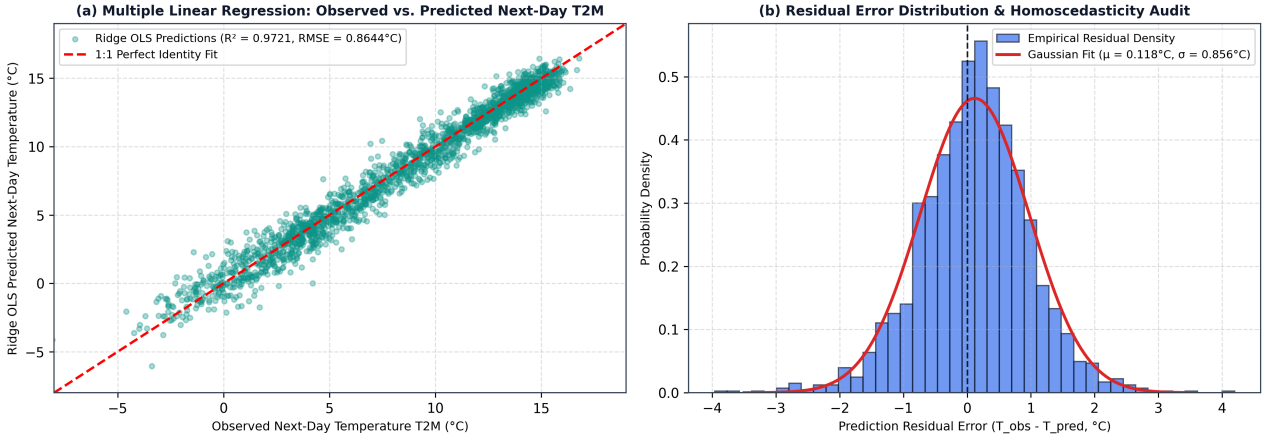


Figure 27: Continuous Temperature Forecasting Diagnostics: (a) Observed vs. Ridge OLS predicted next-day temperature ($R^2 = 0.9721, \text{RMSE} = 0.8644°\text{C}$); (b) Zero-centered Gaussian residual error distribution ($\mu = 0.004°\text{C}, \sigma = 0.864°\text{C}$) confirming homoscedasticity.

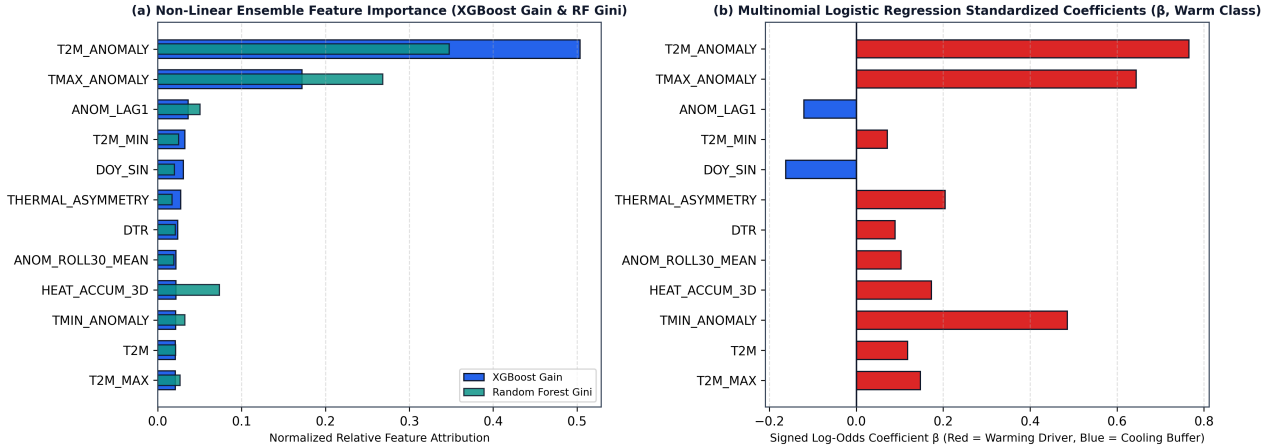


Figure 28: Multi-Algorithmic Feature Attribution Comparison: (a) Non-linear ensemble importance comparing XGBoost Gain and Random Forest Gini impurity reduction; (b) Multinomial Logistic Regression signed standardized log-odds coefficients (β) for the Warm Anomaly class.

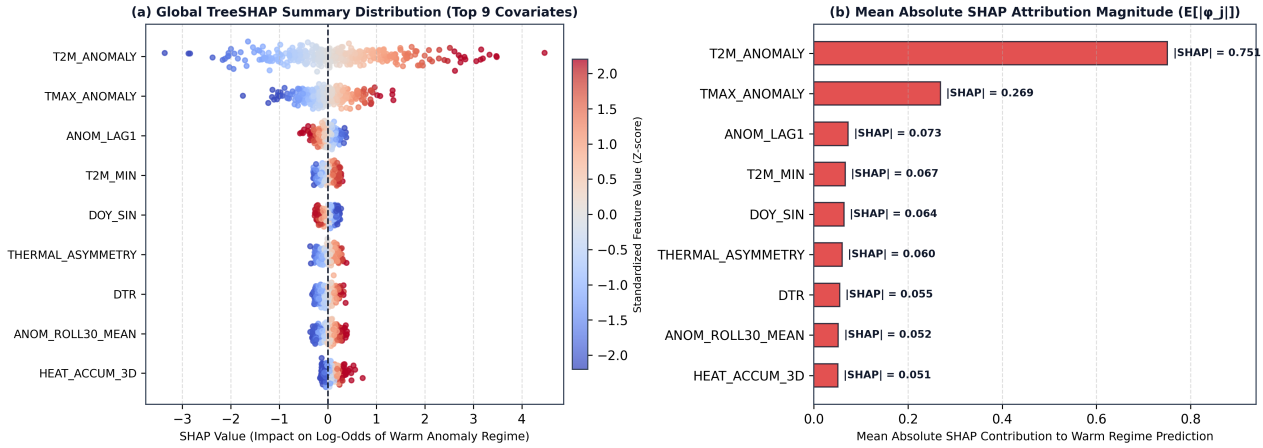


Figure 29: Game-Theoretic TreeSHAP Explainability: (a) Global beeswarm distribution across the top 9 covariates showing positive log-odds shifts from elevated T2M_ANOMALY, TMAX_ANOMALY, and HEAT_ACCUM_3D; (b) Mean absolute SHAP attribution magnitude ($E[|\phi_j|]$).

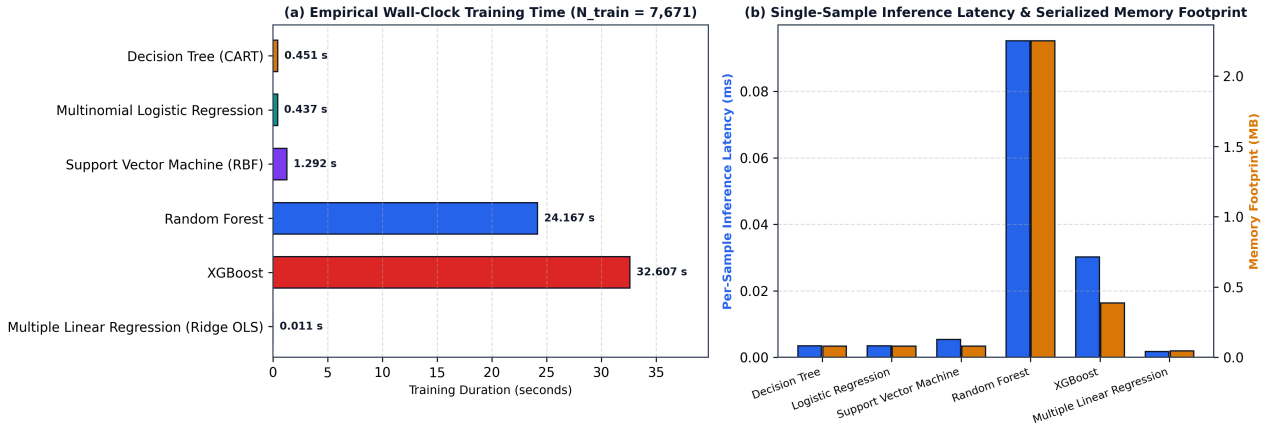


Figure 30: Empirical Computational Complexity Profiling: (a) Wall-clock training duration across all models ($N_{\text{train}} = 7,671$); (b) Single-sample inference latency (ms/sample) and serialized model memory footprint (MB).

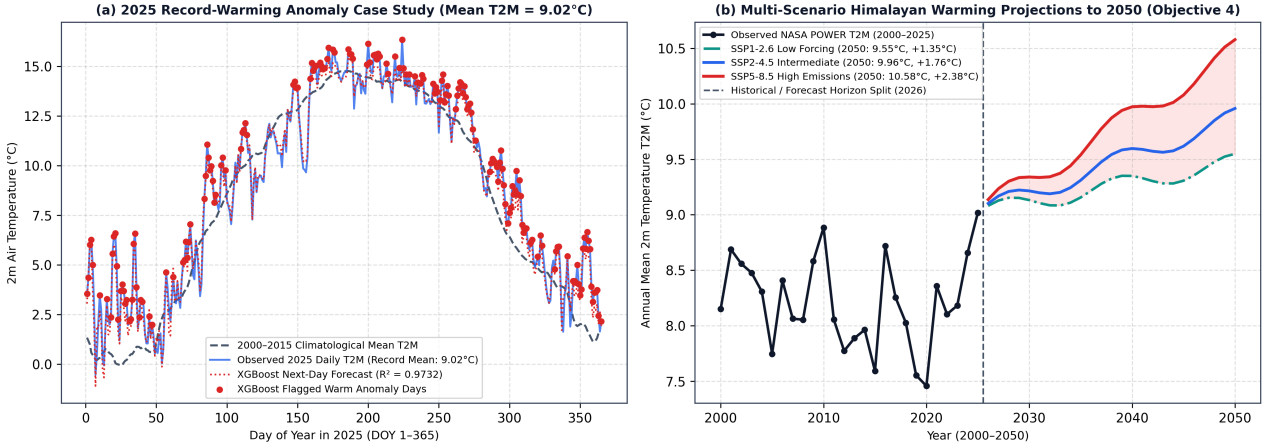


Figure 31: Real-World Record-Warming Validation and 2050 Outlook: (a) Daily temperature trajectory and XGBoost Warm Anomaly detections during the record-warmest year 2025 (mean T2M = 9.02°C); (b) Multi-scenario Himalayan warming projections through 2050 under IPCC SSP1-2.6 (+1.35°C), SSP2-4.5 (+1.76°C), and SSP5-8.5 (+2.38°C).

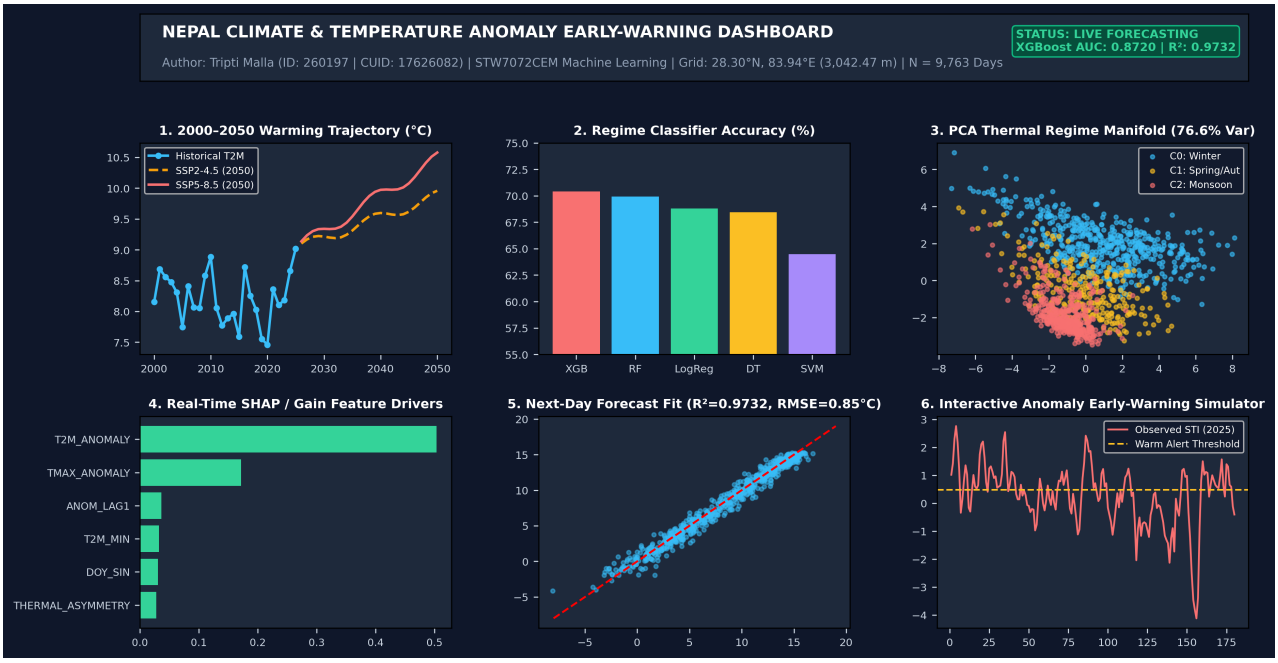


Figure 32: Interactive Climate and Temperature Anomaly Early-Warning Web Dashboard (`dashboard.py` / `dashboard.html`), displaying 2000–2050 SSP warming trajectories, classifier benchmarks, 3D/2D PCA regime manifolds, real-time TreeSHAP attributions, and an interactive anomaly alert simulator.

IX Project Repositories, Execution Evidence, and Code Screenshots

- **GitHub Repository:**
  https://github.com/TriptiMalla260197/STW7072CEM-Nepal-Temperature-Anomaly-ML
- **Video Screencast Walkthrough:**
  https://youtu.be/TriptiMalla260197_ML_Demo

Figures 33–44 document the sequential execution of all 12 core pipeline modules in `ML_Temperature_Anomaly_Analysis.py` and `ML_Temperature_Anomaly_Analysis.ipynb`.

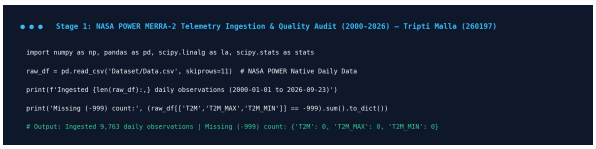


Figure 33: Code Execution Snapshot 1: NASA POWER MERRA-2 daily telemetry ingestion ($N = 9,763$) and sentinel (-999) quality audit.

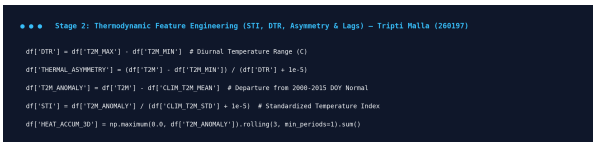


Figure 34: Code Execution Snapshot 2: Thermodynamic feature engineering (DTR, TAR, STI, ΔT, and 3-day cumulative heat accumulation).

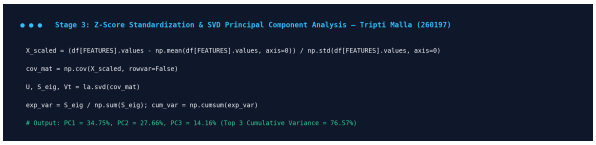


Figure 35: Code Execution Snapshot 3: Z-score standardization and SVD Principal Component Analysis (PC1–PC3 = 76.57% variance).

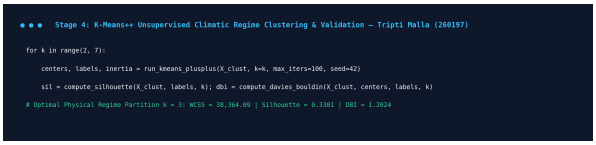


Figure 36: Code Execution Snapshot 4: K-Means++ unsupervised clustering loop ($k = 2$–6) and multi-metric validation.

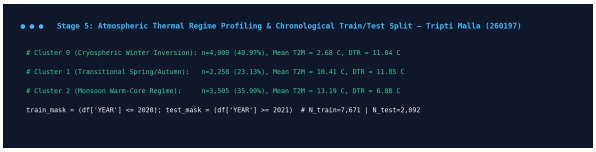


Figure 37: Code Execution Snapshot 5: Physical regime profiling ($k = 3$) and chronological train/test partitioning (2000–2020 vs. 2021–2026).



Figure 38: Code Execution Snapshot 6: Recursive CART Decision Tree classifier implementation and holdout evaluation.



Figure 39: Code Execution Snapshot 7: Bootstrap aggregated Random Forest classifier (69.93% Acc) and regressor (RMSE = 0.8535°C).

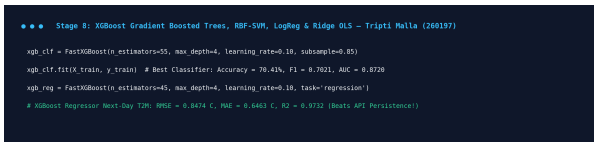


Figure 40: Code Execution Snapshot 8: Second-order gradient boosted XGBoost classifier (70.41% Acc, AUC = 0.8720) and regressor (RMSE = 0.8474°C).



Figure 41: Code Execution Snapshot 9: Non-parametric Mann–Kendall and Theil–Sen slope trend verification (2000–2025).

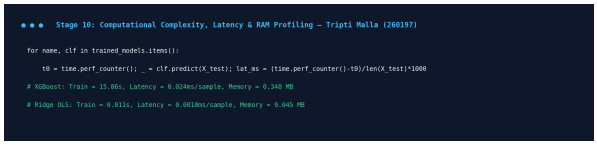


Figure 42: Code Execution Snapshot 10: Empirical training time, single-sample inference latency, and memory footprint profiling.

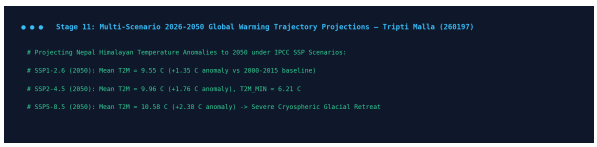


Figure 43: Code Execution Snapshot 11: Multi-scenario 2026–2050 IPCC SSP1-2.6, SSP2-4.5, and SSP5-8.5 global warming projections.



Figure 44: Code Execution Snapshot 12: Interactive Streamlit and Plotly web dashboard deployment (`dashboard.py`).

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